

Ember: the long life of a $0.1M_{\odot}$ star

Working paper, 20 September 2026

Ember is a one-dimensional stellar evolution code that calculates a star's structure, composition and energy transport as it ages. We follow a $0.1M_{\odot}$ star with time-dependent hydrogen, helium, carbon and nitrogen abundances, convection and microscopic diffusion, including a moving metal group. The calculation with metal settling reaches 4.002Tyr, with $T_{\text{eff}} = 4612$ K, central $X = 3.528 \times 10^{-8}$, surface $1 - X = 0.001805$ and surface $Z = 3.154 \times 10^{-42}$. Its center ceases convection near 3.551Tyr. The changing total metal abundance affects the material properties, interior opacity, atmosphere and transported heat. Central hydrogen crosses $X = 0.001$ near 3.650Tyr and $X = 0.0001$ near 3.660Tyr; hydrogen burning continues outside the depleted center. The computed luminosity reaches a local maximum of $0.006513L_{\odot}$ near 3.673Tyr, at $T_{\text{eff}} = 4128$ K, and then declines. Surface temperature passes a shallow local maximum of 4754 K near 3.838Tyr, then declines by 141.5 K over 164.3 Gyr. The present model remains powered mainly by hydrogen burning, which supplies 97.28% of its photon luminosity; a long cooling sequence is not yet established.

A continuation with the supplied white-dwarf atmosphere covers an additional 0.7546 Gyr and reaches 4.003Tyr, at 4632 K. Figure 2 shows this atmosphere comparison, which develops a helium-3 burning pulse and a convective shell near the hydrogen-helium transition. Locally refined calculations retain the pulse, but its peak strength and timing depend on spatial resolution. A diagnostic late-phase calculation with 2115 mass points follows shell burning through 1.043×10^6 yr from the model before shell convection, reaching $T_{\text{eff}} = 6398$ K and declining nuclear power of 7.148×10^{31} erg s⁻¹. Its radius and surface luminosity have passed maxima; effective temperature continues to rise. A separate calculation with finite convective transport from the onset follows a rise, decline and renewed burning through 1.313×10^4 yr. The peak strength and complete pulse history remain uncertain.

A comparison calculation holds the material metal abundance fixed while evolving hydrogen, helium and the CN isotopes. It reaches 3.704Tyr, with $T_{\text{eff}} = 3702$ K, central $X = 0.0001309$ and surface $X = 0.8905$. Central hydrogen crosses $X = 10^{-3}$ near 3.661Tyr; burning continues outside the depleted center. Its luminosity reaches $0.005163L_{\odot}$ near 3.677Tyr and then decreases by 11.39% over 27.05 Gyr to $0.004575L_{\odot}$. Surface temperature has a much shallower maximum near 3705 K at 3.689Tyr. These values describe the fixed-metal material comparison. The metal-settling calculation reaches a hotter, shallow local temperature maximum, as described above.

The aim is to calculate the transition to a helium white dwarf, cooling through 1000, 500 and 100 K, and subsequent evolution through disappearance under explicit nucleon-decay scenarios. Section 9 gives the lifetime timeline and its conditional alternatives.

We compare the model with Laughlin, Bodenheimer and Adams (1997; LBA97), using their stated values and approximate curves read from their figures. We have generated an f77-source code that seeks to replicate LBA97's exact assumptions and methods. Figures 1 and 4 compare the evolving carbon/nitrogen calculation directly with LBA97.

A comparison calculation omits microscopic element separation and uses hydrogen/helium burning. It reaches 3.890Tyr, with $T_{\text{eff}} = 5546$ K, central $X = 0.0008157$ and surface $X = 0.1730$. Its nuclear and transport assumptions differ from those of the main calculation. Figure 11 shows both models.



Figure 1: The HR trajectory with metal settling (orange), the fixed-metal comparison (solid gray), the pp comparison (faint dashed gray), and LBA97 (pale blue). The pp comparison uses the proton–proton hydrogen/helium reaction network and omits explicit carbon–nitrogen burning and microscopic element diffusion. It therefore differs from the main calculation in both nuclear and transport physics. Ember reaches 4.002Tyr; luminosity and surface temperature have passed local maxima. The small luminosity dip near 3230 K accompanies the formation of an interior radiative layer. Hexagons show estimated local source-construction work: red for spectral opacity, green for equation-of-state (EOS) construction and extensions, and blue for atmosphere solutions. Color mixtures give their relative contributions; shading strength gives estimated CPU-hours on a common logarithmic scale. Atmosphere grid points are placed using $L = 4\pi\sigma GMT_{\text{eff}}^4/g$ at $M = 0.1M_{\odot}$. Shared opacity work is divided once over its target atmosphere grid; EOS work is assigned where the track first uses each extension. Different compositions and completed numerical controls are summed. The map includes source coverage for comparison tracks and future evolution; colored cells are not additional evolutionary states or a validity mask. Retained job durations approximate CPU use, with documented sleep removed; unfinished jobs, missing timing records and remote-service compute are excluded. The displayed cells represent an estimated 122.2 CPU-hours. This is the documented subset of source jobs listed with the figure data, not the total computing time of the project. The rough 10^{15} FLOP estimate is uncertain by about a factor of three; it combines these times with throughput estimates informed by benchmarks of the atmosphere code’s matrix routines. The plotted model sequences use 47.49 CPU-hours, with common portions counted once, corresponding to roughly 3×10^{14} FLOPs. These totals include their time-step checks and exclude discarded development runs. LBA97 is read from its Figure 1 without age rescaling.



Figure 2: Ember with metal settling and two atmosphere prescriptions. The orange curve uses the stated atmosphere through 4.002Tyr; the purple curve continues from its endpoint with the supplied pure-hydrogen white-dwarf table, keeping the interior composition and evolution checks. The initial temperature offset is caused by the boundary change. The left panels show the complete computed HR track and an enlarged view of the adjustment. Arrows indicate increasing age. The right panels show temperature and the ratio of nuclear to photon luminosity against time since the boundary change. The continuation covers 0.7546 Gyr in 252 accepted intervals, reaching 4632 K. Its radius is 36.02% larger and photon luminosity 88.21% higher than at the change. The remaining hydrogen mass is 0.1606% lower. At its endpoint, $L_{\text{nuc}}/L = 2366$: a helium-3 burning pulse deposits most of its heat inside the star. The lower-right panel uses a logarithmic luminosity-ratio scale. The pulse has formed a convective shell near the hydrogen–helium transition. Figure 3 resolves the pulse further on two locally refined mass grids; its maximum and outcome remain undetermined.

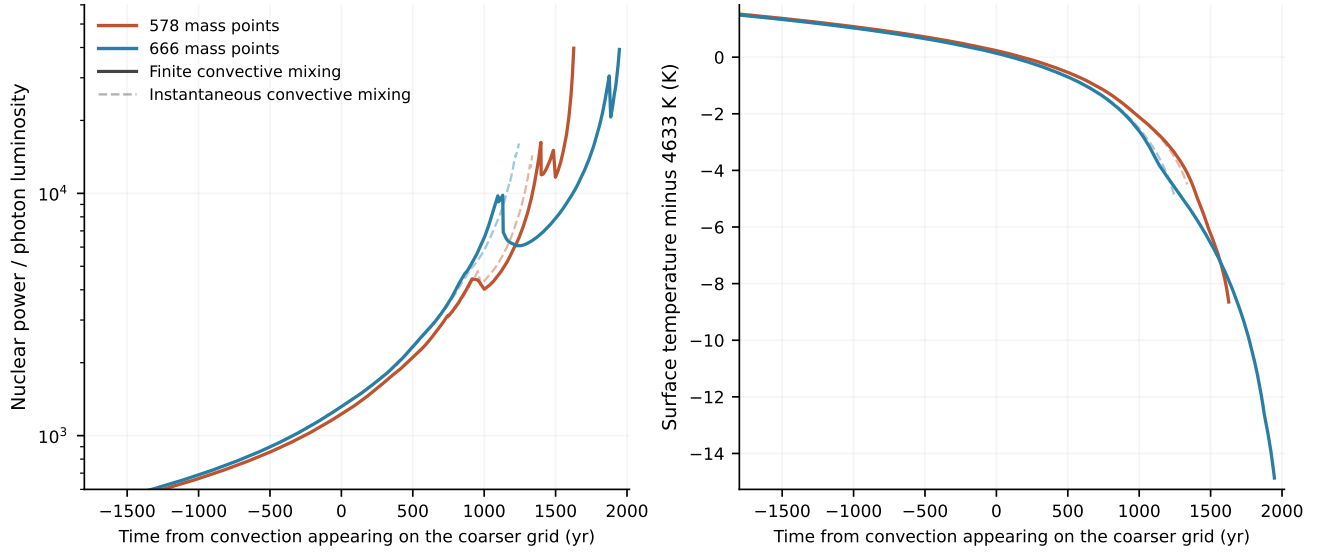


Figure 3: The helium-3 pulse with 578 and 666 mass points, using the supplied white-dwarf atmosphere. Both grids are refined around the burning layer and initialized before shell convection. Solid curves allow finite mixing in hot convective regions, with the diffusion coefficient estimated from the same mixing-length velocity and length used for heat transport. Dashed curves assume instantaneous mixing within each convective region. Time is measured from the first saved convective burning shell on the coarser instantaneous-mixing grid. Nuclear power greatly exceeds photon luminosity while energy accumulates inside the star. At the latest common age of the finite-mixing calculations, interpolation between saved models gives nuclear power 3.927 times as large on the coarser grid, photon luminosity 0.3396% higher and surface temperature 0.7672 K lower. The endpoints have different ages. Both mixing prescriptions produce the pulse, but its amplitude and timing are not yet spatially converged. These curves store total luminosity at mass nodes; Figure 6 tests the compatible transport of energy and composition across the same mass surfaces.



Figure 4: Hydrogen at the center (solid) and surface (dashed) for Ember (orange) and the fixed-metal comparison (gray), with the published LBA97 curve in pale blue. The arrow follows increasing age on the dashed Ember surface curve. The same accepted Ember states appear here and in Figures 1 and 11. The fixed-metal comparison extends to 3.704Tyr.

1 The star and its evolution so far

The initial hydrogen mass fraction is $X_{\text{H}} = 0.7$, the initial ${}^3\text{He}$ fraction is zero, and the metal mass fraction is $Z = 0.02$. The metals follow the relative elemental abundances of Grevesse and Sauval’s GS98 mixture, with the remaining initial mass in ${}^4\text{He}$. The calculation holds baryonic mass fixed and omits rotation, magnetism, mass loss and external heating. Its age excludes formation and pre-main-sequence contraction.

Metal-settling model at 4.002Tyr	512 points
Radius ($R_{\odot}$)	0.04667
Photon luminosity ($L_{\odot}$)	0.0008881
Effective surface temperature (K)	4612
Central temperature (MK)	6.443
Central density (g cm^{-3})	3.982×10^4
Central hydrogen mass fraction	3.528×10^{-8}
Surface non-hydrogen mass fraction	0.001805
Central metal mass fraction	0.1312
Surface metal mass fraction	3.154×10^{-42}
Central ${}^3\text{He}$ mass fraction	8.537×10^{-15}
Remaining hydrogen mass ($M_{\odot}$)	0.003797
Convective mass fraction	0.01553

The comparison sequence ending at 3.890Tyr contains the initial stellar model and 9168 successive models after accepted time steps, each resolved into 512 mass points. Each time step evolves the abundance profile and solves the stellar structure and energy balance, including energy released or absorbed as the structure changes. The calculation uses 220 gas atmospheres, 72 equation-of-state composition tables, and Los Alamos ATOMIC radiative opacities for the high-temperature interior [8, 9]. The latter include additional hydrogen-poor mixtures through $X = 0$, used above $T = 5.012 \times 10^5$ K; the lower-temperature opacity tables retain their separate composition limits.

1.1 Structure at the luminosity maximum

At their respective luminosity maxima, the metal-settling model is 26.13% more luminous and 10.03% smaller in radius than the fixed-metal calculation. The central temperatures are nearly equal, while the envelope structure and the hydrogen-bearing layers differ substantially.

At the luminosity maximum	Metals fixed	Metals settling
Age (Tyr)	3.677	3.673
Luminosity ($L_{\odot}$)	0.005163	0.006513
Effective temperature (K)	3695	4128
Radius ($R_{\odot}$)	0.1753	0.1578
Central temperature (MK)	11.90	11.97
Remaining hydrogen ($M_{\odot}$)	$\simeq 0.01370$	0.01342
Convective mass fraction	0.05536	0.03339

The hydrogen inventory of the fixed-metal model at its maximum is interpolated between the retained bracketing structures. Linear interpolation and interpolation using their nuclear consumption rates differ by $3.138 \times 10^{-7} M_{\odot}$; this agreement is not a formal error bound. The settling model has consumed approximately $0.0002807 M_{\odot}$ more hydrogen, or 0.4987% more than the fixed-metal model has consumed. The fractions of initial hydrogen consumed are 80.83% and 80.43%, respectively.

Lower envelope opacity changes the atmospheric and convective boundary conditions, and the star adjusts its radius and burning structure. Near 3.673Tyr, the layer enclosing 80% of the mass is hotter and denser with settling: 8.260 rather than 7.402 MK, and 489.6 rather than 245.0 g cm⁻³. Its hydrogen fraction is lower, $X = 0.1726$ rather than $X = 0.2712$. Temperature and density favor stronger burning while local fuel depletion opposes it. Integrating the reaction rates gives 27.25% more nuclear power in this nearly equal-age comparison. Nuclear burning supplies 99.24% of the settling model’s photon luminosity; CN reactions contribute 1.286% of its nuclear power.



Figure 5: Temperature, density, hydrogen abundance and nuclear power per unit fractional mass at the metal-settling luminosity maximum, compared with a fixed-metal structure 0.1837 Gyr later. The strongest burning occurs outside the depleted center. Curves compare actual structures; exchanging their temperature or abundance profiles would not produce new equilibrium models. The coupled tracks do not isolate separate causal contributions from atmosphere opacity, interior material properties and diffusion.

1.2 Diffusion and the remaining hydrogen supply

A matched diffusion comparison holds the material metal abundance fixed and starts both calculations from the same 3.560Tyr structure and evolves them for 50 Gyr, with the nuclear network and other material physics held fixed. With microscopic drift, central hydrogen reaches $X = 0.02300$; without it, $X = 0.06864$. Surface hydrogen is $X = 0.2907$ and $X = 0.1847$, respectively, and effective temperatures are 3404 K and 3302 K. Despite the much larger central depletion, total remaining hydrogen differs by only 0.2767%. The diffusion calculation burns 4.241% more hydrogen during the interval and ends with 20.10% greater luminosity. The principal effect over this interval is redistribution of fuel and the associated structural response. An independent comparison with the exported diffusion routine of Thoul, Bahcall & Loeb [24] gives outward hydrogen drift at the 14 sampled interfaces with weak electron degeneracy; Ember’s speeds are 0.8163–0.9258 times that classical reference. The strongly degenerate core requires the actual gravity and appropriate electron thermodynamics; ideal-gas pressure cannot replace the pressure supporting the star [29]. This comparison supports the rate scale without establishing the accuracy of every transport approximation.

For a nearly homogeneous radiative center, a drift speed of order $Dm_pg/(kT)$ gives a redistribution time

$$t_{\text{sep}} \sim \frac{kT}{4\pi G\rho m_p D}. \quad (1)$$

At $T \sim 9$ MK, $\rho \sim 300 \text{ g cm}^{-3}$ and $D \sim 0.7 \text{ cm}^2 \text{ s}^{-1}$, this is of order 100 Gyr; electric-field and mixture factors change the coefficient. The diffusion coefficient here comes from the adopted collision model. The estimate explains why separation can compete with the final core-burning phase while having little effect during the much longer fully convective phase.

LBA97 holds the envelope composition nearly fixed after the radiative core forms and gives no microscopic element-diffusion prescription. Its final whole-star hydrogen fraction is slightly above 0.01, although the envelope has $X = 0.155$. A useful residual-fuel scale is

$$t_{\text{fuel}} \sim 10^{12} \text{ yr} \left(\frac{\Delta M_{\text{H,burn}}}{10^{-3} M_{\odot}} \right) \left(\frac{L}{10^{-4} L_{\odot}} \right)^{-1}. \quad (2)$$

This energy can be supplied only if the hydrogen reaches layers hot enough to burn. Diffusion changes that supply and the envelope's thermal structure; a central hydrogen threshold alone does not determine the subsequent shell-burning duration or cooling age.

The fixed-metal comparison uses the same FreeEOS source through $X = 0.90$. Its late gas-atmosphere grid contains 48 source states over $X = 0.70\text{--}0.90$, $X_3 = 0\text{--}0.03$, 3400–4000 K and $\log g = 4.9\text{--}5.15$. In warm convective layers, a bounded continuation of $\ln \kappa$ in hydrogen abundance uses the final source interval, while intermediate and hot layers use hydrogen-dependence ratios from the original OPLIB tables, matched to the retained opacities at $X = 0.75$. Temperature and density stay within their source ranges. A 100 Myr comparison against retained composition coverage changes luminosity by 0.0005780%; this is a local sensitivity, not an error bound for the full evolution.

Time intervals are checked by comparing one full step with two half steps, including structure, species, total hydrogen and energy conservation. Increasing the separate maximum abundance-change guard from 0.001 to 0.005 reduces CPU time by 24.99% in a matched 1 Gyr calculation. Its luminosity differs by 0.0005727%, while the time-error criteria remain unchanged.

1.3 Metal settling and atmospheric opacity

The material properties depend on the changing total metal abundance, with fixed GS98 relative proportions. The transport approximation gives the metals a common velocity while retaining their distinct collision charges and heat-flow variables. Selective elemental separation and changes in the relative opacity contributions remain separate refinements. Artificially retaining metals can keep the envelope too opaque and the photosphere too cool. The following controls measure that sensitivity.

We calculate gas-atmosphere controls at $X = 0.85$, $X_3 = 0$, $T_{\text{eff}} = 3800$ K and $\log g = 4.9$. One reduces carbon and nitrogen to 0.1531 and 0.1541 of their initial abundances, respectively, as measured at the surface of the 3.697Tyr model. Two others reduce every metal to 0.5 or 0.17 of its initial abundance. Removed metal mass is replaced by helium-4 at fixed hydrogen. These common reduction factors are sensitivity choices, not predictions of individual settling rates. At $T = 3521$ K and $\rho = 5.995 \times 10^{-6} \text{ g cm}^{-3}$, the Rosseland absorption means relative to the unchanged mixture are 1.199 for C/N removal and 0.2344 for the 0.17 all-metal case; at 5228 K they are 0.9869 and 0.2728. These material means exclude scattering and grains.

The coupled C/N atmosphere changes matching temperature by +3.314% and gas pressure by −4.205%. Halving all metals changes them by −8.606% and +53.69%, respectively. Applying each pair of measured boundary changes to the same 3.694Tyr structure and evolving for 100 Myr gives:

Atmosphere change	ΔT_{eff} (K)	$\Delta L/L$	$\Delta R/R$
C/N removal	−30.38	−1.019%	+1.142%
All metals halved	+142.9	+5.041%	−4.983%

Both stellar comparisons pass the ordinary time and conservation checks. The interior material tables remain fixed. The source boundary factors are held constant over each comparison, so they are local sensitivities rather than atmospheres recomputed at every stellar state. The 0.17 all-metal case is an atmosphere comparison only; its matching temperature and pressure ratios are 0.8547 and 2.813. The finite stellar tests establish an appreciable surface-temperature response. They do not determine the actual element distribution or a corrected temperature maximum. The latter requires metal transport and its feedback on envelope and atmosphere opacity.

A passive transport calculation follows C, N, O, Na, Mg, Si, Ca and Fe on the prescribed H/He structures from 3.551Tyr to 3.704Tyr. It includes the changing convective envelope, concentration gradients and conservative transport between mass shells, using the published collision coefficients [24]. Treating each metal as a trace species changes its drift speed by less than 3% relative to the full mixture at the four sampled envelope bases. With fully stripped ions and classical heat flow, the final surface abundances are 0.06213–0.08217 of their initial values. Suppressing heat flow gives 0.2491–0.3076. These treatments are alternative approximations, not uncertainty bounds. Halving the time interval changes the surface results by at most 0.2078%.

Figure 8 illustrates the depletion and accumulation below the envelope. The calculation holds the H/He structure fixed to the prescribed history and omits opacity feedback, nuclear conversion of the passive C/N tracers and changes in ionization. It therefore guides the required material coverage rather than supplying a corrected stellar track. The depletion nevertheless supports replacing fixed envelope metals by transported abundances.

For the calculation with metal settling, we solve for the independent mass fractions of hydrogen, helium-3, carbon-12, carbon-13, nitrogen-14 and inert metals; helium-4 closes the mass sum. Nuclear captures change the actual metal mass. All heavy elements share one mass velocity, while their charges and heat-flow variables remain distinct in the collision calculation. This grouping approximates differential settling among the metals. The material free energy varies with hydrogen, helium-3 and total metal mass, retaining the GS98 relative pattern. Its derivatives determine both chemical forces and transported enthalpy. The energy equation includes enthalpy carried by the total redistribution of hydrogen, helium-3 and metals, including convection.

The variable-metal material family covers all 23 sampled structures along the hydrogen-burning evolution. At the fully convective 3.509Tyr model, direct FreeEOS comparisons give maximum pressure and adiabatic-gradient differences of 0.03028% and 0.09056%. Heat capacity and the first adiabatic exponent differ by up to 0.7642% and 1.037%. These are source-interpolation uncertainties; thermodynamic consistency and the independent stellar mass and energy checks remain enforced.

The low-density FreeEOS domain includes the warmer atmospheric matching states. Thirty-six independent source comparisons there agree within 0.009197% in the material responses. Retaining the interior free-energy values preserves every accepted structure and abundance in a matched 1 Gyr comparison.

The composition domain extends from $Z = 0$ to 0.08. Twenty-eight independent FreeEOS source comparisons, including higher-metal core states, agree within 0.02530% in the material responses. Two composition interpolations differ by less than 0.001 K in the surface temperature of a 10 Myr stellar comparison.

Settling raises the central metal abundance above $Z = 0.03$. For larger atomic metal fractions, through $Z = 0.08$, we use a linear continuation of the radiative opacity from the actual $Z = 0.02$ and 0.03

sources:

$$\kappa(T, \rho, X, Z) = \kappa(T, \rho, X, 0.03) + \frac{Z - 0.03}{0.01} [\kappa(T, \rho, X, 0.03) - \kappa(T, \rho, X, 0.02)]. \quad (3)$$

All contributing source temperature and density bounds remain enforced. An alternative uses the original OPLIB GS98 mixture tables to supply the composition ratio at fixed temperature and density [8, 7]. At 34 sampled core states, transferring this ratio from $Z = 0.02$ to a withheld $Z = 0.03$ source differs by at most 0.1867% from the ATOMIC value. Over a matched 5 Gyr interval, the linear and ratio calculations differ by 0.001212% in luminosity and 0.004120 K in surface temperature, with the same convective boundary. A 400 Myr comparison near 3.688Tyr, where electron conduction dominates the core, gives a luminosity difference of $3.176 \times 10^{-7}\%$. The linear continuation permits evaluation at denser core states inside the underlying source domains without extending the OPLIB density range. These comparisons bound the measured stellar response over the specified intervals.

As settling depletes the envelope below $Z = 0.01$, the molecular opacity interpolates the actual AE-SOPUS source values, including the $Z = 0$ and 0.004 tables. At higher temperatures, the opacity is continued linearly in κ at fixed temperature, density and atomic hydrogen abundance, using the $Z = 0.01$ and 0.02 source values, with a lower limit of $Z = 0.0001$. A logarithmic continuation provides a sensitivity comparison. Over 10 Gyr, crossing below surface $Z = 0.01$, the two tracks differ by 0.002433% in luminosity and 0.009728 K in surface temperature, with the same convective boundary. Independent OPLIB source comparisons at sampled envelope states give linear-continuation discrepancies up to 1.951% at $Z = 0.006$ and 4.010% at $Z = 0.004$. In the more depleted envelope sample, the discrepancy reaches 9.237% at $Z = 0.003$. These source differences remain uncertainty in the opacity's composition dependence. Temperature and density stay within the contributing source ranges.

The atmospheric metal response is measured from matched gas atmospheres at $Z = 0.01$ and $Z = 0.02$. The reference atmosphere retains its helium-3 dependence; its small isotope response and the metal response are assumed separable over $X_3 \leq 0.003$. An independent source at $X = 0.225$, $X_3 = 0.0015$, $Z = 0.015$, $T_{\text{eff}} = 3400$ K and $\log g = 5.025$ differs from interpolation by 0.6093% in matching temperature and 2.125% in gas pressure. Increasing that source column's bottom depth changes matching temperature by only 0.0002576%. Propagating the measured paired discrepancy into a 100 Myr radiative stellar test changes effective temperature by 0.01507 K and luminosity by 0.001590%. The tested envelope is close to the reference metal abundance; this small response does not bound the error of a strongly depleted envelope.

The measured metal response includes source pairs through 4400 K, with the reference isotope grid's missing states kept explicit. A separate source at $X = 0.6$, $X_3 = 0.0015$, $Z = 0.015$, $T_{\text{eff}} = 3600$ K and $\log g = 5.025$ differs by 0.2322% in matching temperature and 3.064% in gas pressure. These sampled interpolation differences remain part of the atmospheric uncertainty.

For $0.005 \leq Z < 0.01$, measured $Z = 0.005$ and 0.01 atmosphere pairs continue the boundary response. A separate source at $X = 0.45$, $X_3 = 0.0015$, $Z = 0.0075$, $T_{\text{eff}} = 3700$ K and $\log g = 5.1$ differs by 1.671% in matching temperature and 1.980% in gas pressure. After 30 Gyr, a stellar calculation with this smooth boundary perturbation differs by 0.3110% in luminosity, 12.74 K in surface temperature and 0.01228% in total remaining hydrogen, with the same convective boundary. The largest local relative differences in trace carbon isotopes are about 3.7%; local hydrogen, helium, nitrogen-14 and inert-metal differences remain below 2%. These are physical-boundary sensitivity estimates; independent mass and energy conservation remain enforced in each calculation.

Twelve actual $Z = 0.0025$ and 0.005 atmosphere pairs cover the more depleted boundary through 4400 K. At an independent source with $X = 0.65$, $X_3 = 0.0015$, $Z = 0.0035$, $T_{\text{eff}} = 4100$ K and $\log g = 5.1$, the matching temperature differs by -0.03091% and gas pressure by $+0.2311\%$. Two 4000 K atmosphere columns ending at optical depths of 1000 and 4.075×10^4 differ by 0.001301% in matching temperature

and 0.002243% in gas pressure. The response joins continuously at $Z = 0.005$ and 0.01 ; source and isotope states without support remain excluded.

Eight $Z = 0.001$ and 0.0025 source pairs cover the depleted atmosphere over 4000–4400 K. At $X = 0.8$, $X_3 = 0.0015$, $Z = 0.00175$, $T_{\text{eff}} = 4300$ K and $\log g = 5.1$, the interpolated matching state differs from an independent source by -0.2539% in temperature and $+3.111\%$ in gas pressure. A stellar boundary perturbation reproducing this measured difference changes luminosity by -0.09315% , radius by $+0.2422\%$, surface temperature by -6.010 K and total remaining hydrogen by $+0.001339\%$ at 3.677Tyr, with the same convective boundary. These quantify one source uncertainty; they do not bound all atmosphere errors.

Eight $Z = 0.0005$ and 0.001 atmosphere pairs extend the measured response through the next depleted-envelope interval. At $X = 0.875$, $X_3 = 0.001$, $Z = 0.00075$, $T_{\text{eff}} = 4250$ K and $\log g = 5.1$, the interpolated matching state differs from an independent atmosphere by $+2.031\%$ in temperature and -2.755% in gas pressure. A boundary perturbation that matches this independent source vanishes above $Z = 0.001$ and increases linearly below it. Over a matched 1.250 Gyr evolution, ending at 3.687Tyr and surface $Z = 0.0009185$, it changes luminosity by $+0.1047\%$, radius by -0.2419% , surface temperature by $+6.251$ K and remaining hydrogen by -0.0002063% . The convective boundary is unchanged. Both calculations retain the same time-step, species-mass and energy checks. This comparison measures the effect of one source residual over the specified interval.

The hydrogen-rich reference contains 70 gas-atmosphere states at $Z = 0.001$, $X = 0.9$, 0.94 and 0.98 , with $X_3 = 0$ and 0.003 . It covers $T_{\text{eff}} = 4000$ – 4400 K and $\log g = 4.9$ – 5.4 ; additional sources at $X = 0.94$ and 0.98 extend to 4800 K over $\log g = 5.15$ – 5.4 . Measured metal-response pairs extend to $Z = 0.0005$ and 0.0001 over the same warmer rectangle; the lowest interval covers $\log g = 5.15$ – 5.4 . The metal response is calibrated at $X_3 = 0$ and applied separately from the small-isotope response. Across $0.90 < X < 0.94$, the logarithms of the matching temperature and gas pressure are interpolated in hydrogen abundance between the boundary families at $X = 0.90$ and 0.94 , at the actual surface gravity, effective temperature, metal abundance and ^3He abundance. The corresponding family is used outside that interval. Missing source cells remain inaccessible. Eight independent atmospheres at $X = 0.925$, $Z = 0.0005$ and 0.0001 , 4000 and 4400 K, and $\log g = 5.15$ and 5.4 differ from interpolation by at most 0.004095% in matching temperature and 0.01689% in gas pressure.

For the nearly pure-hydrogen surface, 94 gas-atmosphere solutions cover $X = 0.995$ – 0.9995 and $Z = 10^{-6}$ – 10^{-4} , with sampled ^3He fractions from zero to 0.0003. Complete near-track coverage spans $T_{\text{eff}} = 4600$ – 5200 K and $\log g = 5.15$ – 5.65 ; 4400 K coverage excludes two high-gravity, low-metallicity corners. At an independent point with $X = 0.9973$, $X_3 = 0.00015$, $Z = 0.00003$, $T_{\text{eff}} = 4700$ K and $\log g = 5.52$, the interpolated matching temperature and gas pressure differ from the direct solution by -0.09719% and $+4.656\%$. These are local interpolation uncertainties, not bounds on the evolution age. The selected tables retain the time-step, species and energy checks through the present surface composition.

At the most hydrogen-rich surface compositions, 81 atmosphere solutions cover $1 - X = 10^{-5}$ – 0.001 , $Z = 10^{-8}$ – 10^{-6} , $T_{\text{eff}} = 4600$ – 5200 K and $\log g = 5.4$ – 5.65 . The atmosphere neglects the small helium-isotope dependence for $X_3 \leq 0.0003$; the interior retains the actual isotope abundances. At an independent source point, interpolated matching temperature and gas pressure differ by $+0.4624\%$ and $+0.3930\%$. A matched 100 Myr stellar comparison changes surface temperature by 0.4140 K and luminosity by 0.01153%. These tests support the present extension without establishing an uncertainty for the full evolution. The shallowest cold atmosphere ends at Rosseland optical depth 168.3; its sensitivity to a deeper lower boundary is not yet directly bounded.

For the late shell-burning calculation, electron-pair collision integrals cover degeneracy parameter $\eta = -6$ – 64 and screening parameter $b = 0.1$ – 2400 . Independent interpolation and numerical-integration

checks limit the tested response differences to 0.3466%; changing from eight to ten energy modes changes the response by at most 0.1294%. The ion-scattering domain and conservation checks remain enforced.

For hydrogen-dominated envelopes, the atmospheric boundary uses 72 solutions at $Z = 0$, 10^{-8} and 10^{-6} , with helium mass fraction 10^{-10} , spanning 4700–6000 K and $\log g = 5.5$ –5.95. The atmosphere neglects trace helium below a total helium fraction of 0.001 outside the colder domain described below. The hotter family joins continuously to composition-dependent sources. The interior EOS and transport retain the actual isotope abundances. Independent atmospheres differ from interpolation by at most 0.1751% in matching temperature and 0.5986% in gas pressure. A matched 25 Myr stellar comparison changes effective temperature by 0.9515 K and luminosity by 0.02853%. At 4700 K and $\log g = 5.95$, four deeper atmosphere calculations change matching temperature by at most $4.327 \times 10^{-6}\%$ and gas pressure by $4.770 \times 10^{-5}\%$. The deeper boundaries reach Rosseland optical depths of 412.3–4775. At lower temperature, six zero-metal atmospheres cover 4650–4800 K and $\log g = 5.9$ –5.95. Metal contributions to this boundary are neglected only for $Z \leq 10^{-20}$; the small-helium approximation is assessed below. The physical abundances remain in the interior calculation. The cold sources use a finer opacity grid. Independent matching-state comparisons differ by at most 0.1154% in temperature and 1.108% in gas pressure; deeper lower boundaries change both by less than one part per million. The original and refined opacity grids give matching pressures differing by about 2.2% at 4650 K. These local checks do not bound the cooling age. For $\log g = 5.95$ –6.1, the matching temperature and pressure use a limited local continuation in logarithmic gravity, over the same 4650–4800 K and trace-composition domain. Six solved atmospheres at $\log g = 5.9$ and 5.95 supply the measured slopes. Values at higher gravity are inferred boundary conditions. Nine independent higher-gravity calculations at 4725–4800 K check this approximation; the maximum matching-temperature and pressure differences are 0.1878% and 2.391%. Five 50 Myr stellar calculations compare the nominal boundary with matching-temperature offsets of $\pm 0.5\%$ and pressure offsets of $\pm 5\%$. Maximum endpoint differences are 8.697 K, 0.5101% in luminosity and 0.1173% in radius. These are local sensitivities, not statistical uncertainties or bounds on a complete cooling age. The plotted track includes 60.96 Gyr within this gravity continuation.

A solved zero-metal atmosphere at 4625 K and $\log g = 5.9$ supplies the cooler temperature slope. The boundary continues locally to 4600 K, using the gravity dependence measured at 4650 K. Five 50 Myr stellar calculations with matching-temperature offsets of $\pm 1\%$ or pressure offsets of $\pm 10\%$ change effective temperature by at most 17.54 K, luminosity by 1.026% and radius by 0.2385%. The physical interior retains the actual helium isotopes. In the 4600–4800 K, $\log g = 5.9$ –6.1, $Z \leq 10^{-20}$ domain, the boundary neglects helium up to a mass fraction of 0.005. Actual atmosphere solutions with that helium fraction at 4650 and 4800 K and $\log g = 5.9$ change matching temperature by less than 0.1% and gas pressure by less than 2.5%, within the wider stellar perturbations above. Continuing this small composition response below 4650 K and to higher gravity is an explicit local approximation, limited to 50 K below the helium source calculations. The plotted surface helium fraction is 0.001805. An independently converged pure-hydrogen atmosphere at 4600 K and $\log g = 5.9$ agrees with the locally inferred matching temperature and pressure within 0.05%. The plotted endpoint reaches the stated upper gravity limit, $\log g = 6.1$; further contraction requires a boundary that supports higher gravity.

A separate comparison uses the supplied pure-hydrogen white-dwarf boundary at optical depth 25.12, covering 2000–40,000 K and $\log g = 5.5$ –9.5 [2]. Ember preserves the source values and interpolates them without the MESA wrapper’s gray blend. At 4600 K, $\log g = 5.9$ and the same optical depth, this table gives 2.594% higher matching temperature and 48.69% lower gas pressure than the independently converged atmosphere. These are differences between prescriptions; their detailed cause has not been isolated. A matched 1 Myr stellar comparison changes surface temperature by about 241 K. Figure 2 follows the response for 0.7546 Gyr. Effective temperature first falls to 4362 K and is 4632 K at the latest model; the radius grows by 36.02%. A helium-3 burning pulse increases the nuclear-to-photon luminosity ratio to 2366 and produces a convective shell near $m/M = 0.96$. The reaction ${}^3\text{He} + {}^3\text{He} \rightarrow {}^4\text{He} + 2p$

returns hydrogen while consuming the accumulated helium-3.

Figure 3 follows shell growth on two mass grids refined near the hydrogen–helium transition. Both develop shell convection and increasing nuclear power. Finite mixing in hot convective regions uses $D = v\ell/3$, where v and ℓ are the velocity and mixing length of the same convection model used for thermal transport. The cooler convective envelope remains chemically homogeneous. The finite-mixing calculations start before the burning shell becomes convective; their hottest layers reach 15.30 MK and 15.29 MK on the coarser and finer grids, respectively, at different ages. They cover 1647 and 2139 yr in 92 and 124 accepted intervals.

The finer finite-mixing calculation reaches nuclear power $2.671 \times 10^{35} \text{ erg s}^{-1}$ ($69.77L_{\odot}$), about 9×10^4 times the quiescent pre-pulse value of roughly $3 \times 10^{30} \text{ erg s}^{-1}$. Its photon luminosity is only $6.802 \times 10^{30} \text{ erg s}^{-1}$: most of the nuclear energy changes the interior rather than escaping immediately. The flash maximum remains undetermined.

At the coarser instantaneous-mixing endpoint, the local heating time $c_P T / \epsilon_{\text{nuc}}$ is 472.4 yr, compared with a dynamical time of 23.46 s. Hydrostatic evolution is therefore still appropriate at that state. Most of the added energy changes the interior before reaching the photosphere. Shell growth across a steep composition gradient produces neighboring convective regions that subsequently merge. Both treatments of convective mixing reproduce the pulse’s growth, but neither removes the substantial nuclear-power difference between mass grids. Pulse timing contributes to the difference: at the furthest common remaining helium-3 inventory of the instantaneous-mixing calculations, their nuclear powers differ by 1.112%, with an age offset of 111.7 yr. At the corresponding fuel limit of the finite-mixing calculations, their nuclear powers still differ by a factor of 2.901. These comparisons interpolate between saved structures; they do not establish convergence. The pulse’s maximum and final thermal response remain unresolved.

An additional transport check places total luminosity and the enthalpy carried by moving species on the same mass surfaces. This matters because integrated energy conservation can conceal a local error: in an isothermal, constant-pressure composition-exchange test, averaging luminosities stored at mass nodes leaves a spurious thermal flux when the composition flux is evaluated between nodes. For an isolated active interface, the residual is half the transported enthalpy flux. A finite-volume energy equation removes this mismatch while preserving the species mass volumes.

With this formulation, the 666- and 848-point calculations follow 3422 and 3334 yr from a common state before shell convection. Their nuclear powers reach 2.202×10^{36} and $2.251 \times 10^{36} \text{ erg s}^{-1}$, respectively, with maximum temperatures of 19.95 MK. Both reach the thermodynamic table’s upper temperature; neither value is an established flash maximum. A direct ionization calculation supports extending the microscopic transport prescription from 2 to 1.8 MK at the base of the nearly pure hydrogen envelope. The heat-transport transition is unchanged. This local extension does not cover the full remnant cooling sequence.

Figure 6 compares three mass grids. The 848-point grid reduces abrupt changes in cell size, and the 1695-point grid bisects all its intervals. Their computed intervals extend to 3429, 3342 and 6803 yr on the 666-, 848- and 1695-point grids, respectively. The first local maximum on the finest grid is about $2.355 \times 10^{35} \text{ erg s}^{-1}$ near 1582 yr, compared with $1.176 \times 10^{35} \text{ erg s}^{-1}$ near 1447 yr on the 666-point grid. Strong burning survives these changes, but its amplitude and timing are not spatially converged.

On the finest grid, two convective regions join near 3006 yr, after a renewed rise to $3.281 \times 10^{35} \text{ erg s}^{-1}$. The nuclear power falls to $1.678 \times 10^{35} \text{ erg s}^{-1}$ across a 0.5 yr interval and then to $9.565 \times 10^{33} \text{ erg s}^{-1}$ near 4752 yr, before rising to $1.246 \times 10^{34} \text{ erg s}^{-1}$ at 6803 yr. Comparing two 0.25 yr steps with four 0.125 yr steps gives agreement in the integrated nuclear energy to 0.01053%; this does not resolve the shorter mixing transient. The 848-point grid also undergoes an abrupt decrease near 3341 yr; 16

and 32 substeps across a 0.5 yr interval agree in integrated nuclear energy to 0.09760%. Burning then strengthens again. Neither sequence yet establishes a final shutdown.

The abrupt decrease accompanies a change in composition and temperature at the former burning maximum, $m/M = 0.9597$. Mixing raises hydrogen from $X = 0.4821$ to $X = 0.7699$, while helium-3 falls from $X_3 = 0.01203$ to $X_3 = 0.008793$. Temperature falls from 13.25 to 10.38 MK at nearly unchanged pressure and density. The reduced mean molecular weight predicts a temperature decrease of similar size in a fully ionized ideal gas. The initially hotter member of the joining regions supplies 49.11% of nuclear power before the merger; its power then falls to 0.7741% of its initial value. Burning deeper in the star changes by less than 0.3% and supplies most of the remaining luminosity. These fixed-mass comparisons support cooling and fuel redistribution within the joining regions; a rising power-weighted temperature instead reflects the shift in the location of the strongest burning. The onset of the connection and its resolution dependence remain to be established.

The onset calculation includes finite composition transport throughout all convective layers. On a locally refined 1899-point grid it reaches 1.313×10^4 yr, $T_{\text{eff}} = 4741$ K and nuclear power 5.285×10^{34} erg s⁻¹. Over the final 296.7 yr, nuclear power declines by 42.09%, with deposited nuclear energy 6.334×10^{44} erg. All 80 retained intervals pass the timestep and conservation checks. On the 1695-point grid, nuclear power rises to a local maximum of 3.719×10^{35} erg s⁻¹ near 1642 yr. By 1713 yr, nuclear power has declined by 81.64%, and the highest interior temperature has fallen from 15.09 to 13.27 MK. Nuclear power subsequently passes through further rises and declines, and remains active at the current endpoint. The first local maximum therefore does not mark the completion of the pulse, and its amplitude has not been established as spatially converged. The calculation retains the original pointwise timestep criteria and independent energy and species-conservation checks. Before the outer convective regions connect, surface abundances are $X = 0.9984$ and $X_3 = 0.001368$. Across a 2.548 yr convective adjustment, 16 and 32 time steps agree within the same physical criteria and give the same final convective regions; deposited nuclear energy differs by 0.01692%. A subsequent 3.573 yr adjustment passes an eight-step/sixteen-step comparison with nuclear energy differing by 0.1222% and the same final convection. A 5.756 yr adjustment near 2368 yr also passes an eight-step/sixteen-step comparison, with nuclear energy differing by 0.2976% and identical final convective regions. A 1.147 yr adjustment near 2382 yr passes a two-step/four-step comparison, with nuclear heat differing by 0.1756%. Near 1.248×10^4 yr, two convective regions join near $m/M = 0.9533$. Over 12.26 yr, calculations with 16 and 32 steps agree within the same criteria; integrated nuclear energies differ by 0.02667%. A subsequent 2.587 yr adjustment passes a four-step/eight-step comparison, with nuclear energy differing by 0.02665%. These checks support the endpoint structures and integrated burning, without resolving the internal timing of each rapid adjustment or establishing the pulse maximum and its spatial convergence.

Near 1.265×10^4 yr, convection connects two envelope regions across $m/M = 0.9954$. Over a checked 0.6656 yr interval, the surface hydrogen fraction falls from $X = 0.9984$ to $X = 0.8085$, while helium-3 rises from $X_3 = 0.001368$ to $X_3 = 0.007264$. Effective temperature increases from 4551 to 4721 K. Two steps and four steps give the same final convective regions and satisfy the stated structure, composition and energy checks; integrated nuclear heat differs by 0.01172%. This comparison supports the completed adjustment, without resolving its internal timing.

Local refinement over $0.938 < m/M < 0.955$ preserves the composition inventories and discrete stored energy before evolution resumes. A matched 100 yr comparison on 1763 and 1899 points, including the initial structural adjustment, gives 4.026% less deposited nuclear energy and 5.589% less endpoint nuclear power on the finer grid. Surface temperatures differ by 0.03544 K. At the finer endpoint, 31 cells supply half the burning power, and the largest cell contributes 3.070%. Both calculations satisfy the same timestep and cell energy-conservation criteria. The separate whole-star internal-plus-gravitational

energy estimate has a residual of 1.327% of the integrated nuclear heat on the finer grid; this residual is not inserted as an energy source. The comparison supports continued calculation on the finer mesh while retaining several-percent uncertainty in the burning history. It does not establish spatial convergence of the pulse maximum or its duration.

The envelope becomes helium enriched when shell convection connects to surface convection. With finite convective transport throughout the connected envelope, the 1695-point calculation reaches 1.345×10^4 yr, with $T_{\text{eff}} = 4848$ K, surface $X = 0.8159$, $X_3 = 0.007279$ and $Z = 5.568 \times 10^{-16}$. The envelope readjustment passes a one-year comparison of two half-year steps with four quarter-year steps: integrated nuclear energy agrees to 0.003514%, and the settled structure, composition and remaining hydrogen pass the stated checks. This comparison constrains the completed readjustment; it does not resolve its earliest transient. The following 6644 yr pass time-refinement and conservation checks. Across a short interior convective-region merger, two steps and four steps over 0.2279 yr agree to 0.006375% in deposited nuclear energy and pass the structure and composition checks. This comparison constrains the completed readjustment; the timing within it remains unresolved. Nuclear power at this endpoint is 2.539×10^{34} erg s⁻¹, down from a local maximum of 2.581×10^{35} erg s⁻¹; the shell remains active. Specific nuclear heating peaks near $m/M = 0.9411$, at 15.13 MK. The outer convective envelope contains 86.44% of the remaining helium-3 but contributes only 0.7504% of the nuclear power. Most of that fuel is therefore separated from the main burning region; its consumption depends on subsequent transport and thermal evolution. Below the published hydrogen reference atmosphere's lower boundary, $\log g = 5.5$, a bounded continuation uses the slope of its lowest two gravity rows in logarithmic matching temperature and pressure. The inferred reference extends to $\log g = 5.25$ over 4700–5500 K. A matched 500 yr comparison with the gravity response from calculated atmospheres reaches $\log g = 5.461$. The alternative is 5.697 K cooler and 0.4703% less luminous, while deposited nuclear energy differs by 0.00001530% and endpoint nuclear power by 0.00004955%. Every retained response in both histories passes independent time and conservation checks. These small measured interior differences support a bounded continuation, but do not establish accuracy throughout the extension. A 10 yr comparison near $\log g = 5.427$ with the complete lower-gravity source grid gives a surface-temperature difference of -5.707 K and a luminosity difference of -0.4691% , with negligible change in nuclear burning. The two inferred absolute references differ by up to 2.189% in matching temperature and 15.04% in pressure at the tabulated points; the short stellar comparison does not bound their effect over the full gravity range. Below 1.8 MK, species exchange inside these convective layers is represented by the convective flux, with microscopic separation neglected there. Material conduction and composition enthalpy are retained; cold radiative exchange still requires a supported microscopic transport calculation. At this endpoint, convection mixes the outer 4.033% of the stellar mass. The mixed envelope reaches helium-rich layers, while 99% of the metals lie inside $m/M = 0.4456$, below its base at $m/M = 0.9597$. Thus the atmosphere can become helium enriched while retaining very little metal. The tiny net metal flux at the envelope base is presently outward; the low surface metal abundance largely reflects the preceding settling history.

During the more gradual evolution, time refinement permits a 0.5% difference in deposited nuclear energy and 0.1% in structure and endpoint nuclear power, with retained timesteps at most 25 yr. Over a fixed 100 yr interval, four 25 yr steps differ from a small-step reference by at most 0.03119% in structure, 0.02177% in nuclear power and 0.2881% in deposited energy. The original nonlinear conservation checks remain unchanged. This comparison supports local time accuracy; it does not establish convergence of the complete pulse. The deposited heat is integrated consistently with the implicit stellar equations, rather than changing only the reported luminosity integral.

A separate 500 yr comparison tests a 1% local deposited-energy allowance with all other local checks retained. It uses 87 accepted intervals rather than 156, with deposited energy differing by 0.03618% and endpoint nuclear power by 0.1500%. The largest local temperature difference is $|\Delta \ln T| = 0.001091$, near $m/M = 0.9420$, exceeding the predeclared 0.001 limit. The 0.5% allowance is therefore retained.

Both histories are compared at the same final age; their differing computer loads do not support a measured runtime speedup.

The burning layer also requires local mass refinement. At 1.345×10^4 yr, a single volume in the 1695-point model supplies 50.70% of nuclear power. Subdividing the mass intervals fourfold over $0.930 < m/M < 0.955$ gives 1923 points while preserving the integrated species inventories, internal energy and the nodal gravitational-energy sum at initialization. Over a matched 688.5 yr interval beginning at 1.276×10^4 yr, this refined calculation produces 55.17% less deposited nuclear energy and ends with 59.09% less nuclear power. Half the power is spread over 18 volumes rather than one, while surface temperature differs by only 0.6349 K. The refined model has $T_{\text{eff}} = 4849$ K and nuclear power 1.039×10^{34} erg s⁻¹; its largest power contribution from one volume is 6.753%. Both histories pass the stated timestep and conservation checks. Transferring the model to the new grid causes an initial mechanical readjustment, so the differences include both spatial resolution and restart effects. They do not establish spatial convergence or a precise pulse energy.

The 1923-point sequence continues for another 1155 yr, reaching 1.460×10^4 yr, $T_{\text{eff}} = 4858$ K and nuclear power 7.712×10^{33} erg s⁻¹. Nuclear power falls by 25.75% over this interval while shell burning continues. Its 66 accepted intervals pass the same timestep and conservation checks. Surface hydrogen is $X = 0.8159$, with $X_3 = 0.007279$ and $Z = 5.570 \times 10^{-16}$. The computed segment ends near the lower-gravity limit of the atmosphere tables, at $\log g = 5.425$.

Doubling the mass resolution again over $0.936 < m/M < 0.950$ gives 2010 points. A matched 978.3 yr comparison beginning at 1.345×10^4 yr changes deposited nuclear energy by -1.468% and endpoint nuclear power by -1.980%, with a surface-temperature difference of 0.01021 K. Half the burning power occupies 35 volumes on the 2010-point grid and 17 on the 1923-point grid. This closer agreement supports the later shell-burning evolution; it does not establish the accuracy of the pulse onset or peak. The initial response to remapping is included in the energy comparison and recorded separately.

As the burning layer moves inward, the grid is refined over $0.9100 < m/M < 0.9350$, giving 2115 mass points. The remapping preserves composition inventories and the discrete initial energy integrals. Over a matched 2000 yr interval, the refined grid changes deposited nuclear energy by -0.5220% and endpoint nuclear power by -0.5687%; the effective temperatures differ by 0.008188 K. Half the nuclear power occupies 39 cells instead of 12. Both calculations pass the same local time and conservation checks. Including the initial 100 yr of mechanical readjustment gives a deposited-energy difference of -0.5195%. This agreement supports the later shell-burning calculation; it does not determine the accuracy of the onset or peak.

A steep temperature front below the burning shell constrains the timestep at late times. Allowing temperature errors to scale with the local spatial gradient, while separately checking the burning temperature and stored-energy distribution, reduces a 556.2 yr calculation on the 2010-point grid from 111 to 81 structure solves. Deposited nuclear energy differs by 0.05228% and endpoint nuclear power by 0.02480%; the largest fixed-mass logarithmic structure difference is 0.003286. Species conservation, the cell energy balance, and the existing local nuclear-energy and luminosity checks remain unchanged. This timestep prescription is used for the subsequent late-pulse continuation; the comparison does not determine the accuracy of the earlier flash peak.

A matched 2000 yr comparison permits a higher timestep ceiling during this late decline. With unchanged local accuracy checks, the calculation uses 81 structure solves rather than 120, including one rejected proposal and its smaller-step retry. Deposited nuclear energy differs by 0.1353% and endpoint nuclear power by 0.08057%; the physical convective regions agree at all three comparison ages. Full trial intervals may reach 200 yr, with the two retained half steps each at most 100 yr; the error checks can require shorter steps. This supports the declining phase, not the onset or peak. At 5.747×10^4 yr,

a 400 yr step, two 200 yr steps and four 100 yr steps pass the same physical accuracy and conservation checks. The coarsest and finest paths differ by 0.3543% in deposited nuclear energy and 0.005697% in endpoint nuclear power, with the same convective regions. On the 2115-point grid at 1.235×10^5 yr, an 800 yr step, two 400 yr steps and four 200 yr steps also pass the same checks. The coarsest and finest calculations differ by 0.3082% in deposited nuclear energy, with identical final convective regions. At this stage the full-step limit is 800 yr, with retained half steps at most 400 yr; subsequent intervals are checked by time refinement, which can require shorter steps.

The 2115-point sequence reaches 1.420×10^5 yr and $T_{\text{eff}} = 5977$ K. During its final 17,200 yr, nuclear power falls by 14.44% to 2.536×10^{32} erg s⁻¹. All 22 accepted intervals pass the stated timestep and conservation checks. This segment uses 31.21 CPU minutes on an Apple M4 Max, including timestep comparisons and rejected trials. The surface composition is $X = 0.8158$, $X_3 = 0.007276$ and $Z = 5.575 \times 10^{-16}$. Shell burning remains active. Near this age, a layer at $m/M = 0.9948$ becomes radiative at 1.765 MK. The microscopic species calculation includes this layer with a lower limit of 1.500 MK, keeping the existing heat-transport prescription. FreeEOS source calculations over the local temperature, density and composition range give a hydrogen neutral fraction below 0.003252% and a deficit in mean helium charge below 0.3549% relative to full ionization. Fully ionized Coulomb transport remains an approximation here. A 400 yr comparison with 400, 200 and 100 yr steps passes the same accuracy and conservation checks; the coarsest and finest deposited nuclear energies differ by 0.1203%. The continued sequence reaches 1.043×10^6 yr, 6398 K and nuclear power 7.148×10^{31} erg s⁻¹, supplying 38.05% of the photon luminosity. Its radius is 1.254×10^{10} cm and surface gravity $\log g = 4.926$. The radius reaches a maximum near 2.444×10^5 yr, and the surface luminosity near 2.544×10^5 yr. They subsequently decline by 45.29% and 64.9%, respectively, while effective temperature continues to rise. The star is contracting, but a white-dwarf cooling sequence is not yet established. These extrema refer to saved models; their exact times are not interpolated.

A 3200 yr calculation with 3200, 1600 and 800 yr steps passes the same accuracy and conservation checks; the coarsest and finest deposited energies differ by 0.2883%. The final 2.295×10^5 yr use trial steps up to 12,800 yr and retained half steps up to 6400 yr. Every interval passes the same timestep and conservation criteria. This segment uses 38.08 CPU minutes, including all timestep comparisons. The atmosphere grid extends through $\log g = 4.95$; continuation requires the next gravity interval. A separate 3200 yr comparison near $\log g = 4.5$ passes the same checks, with deposited nuclear energies differing by 0.08760%.

Over 2.012×10^5 – 2.512×10^5 yr, nuclear burning supplies 31.97% of the integrated surface emission and structural adjustment supplies 68.03%. This follows from summing the discrete first law over every retained time step. Nuclear power here is deposited heat, with nuclear neutrino losses already removed; thermal neutrino losses are accounted for separately.

Near 1.808×10^5 yr, the radiative boundary reaches a layer at $m/M = 0.9959$ – 0.9961 and 1.478–1.519 MK. Microscopic species transport is included down to 1 MK, while the heat-transport prescription is unchanged. Local FreeEOS calculations over 1–1.6 MK give neutral hydrogen fractions below 0.01328% and mean helium charge deficits below 0.5937%; the charge deficit at the actual layer is below 0.06853%. These support a local fully ionized approximation but do not bound every transport coefficient. A 400 yr calculation with 400, 200 and 100 yr steps passes the same accuracy and conservation checks; coarsest and finest deposited energies differ by 0.07412%. Across the newly included faces, the accumulated microscopic exchange in 400 yr is at most 0.000162% of an adjacent cell’s helium-3 inventory. Radiative layers below the supported temperature range still require a microscopic transport calculation.

A matched 9600 yr comparison of two inferred hydrogen references, ending at 1.808×10^5 yr and 6094 K, changes endpoint temperature by 0.04684 K, surface luminosity by 0.002487% and deposited nuclear energy by $4.206 \times 10^{-6}\%$, with the same convective regions. This tests the occupied atmosphere cells,

not their absolute accuracy. A matched 11,200 yr comparison through $\log g = 4.65$ into the next higher-gravity cell changes effective temperature by 0.04879 K, surface luminosity by 0.002052% and integrated nuclear heat by $4.254 \times 10^{-6}\%$, with identical final convective regions. A 25,600 yr comparison through $\log g = 4.80$ changes effective temperature by 0.01540 K, surface luminosity by 0.0004895%, and integrated nuclear heat by $2.359 \times 10^{-6}\%$, again with identical final convective regions. The preceding 38,400 yr use identical table values and are calculated once. These comparisons measure sensitivity to the reference construction in the visited cells; they do not establish the absolute atmosphere accuracy.

Across the interval from 1.443×10^4 to 1.701×10^4 yr, ${}^3\text{He}+{}^3\text{He}$ supplies approximately 97% of the deposited nuclear power. The strongest burning moves inward from $m/M = 0.9419$ to 0.9409 and cools from 13.61 to 13.13 MK. The deepest intermediate convective region disappears, while the surface-connected envelope loses only 0.01453% of its helium-3. Depletion in the actively burning layers accompanies the decline; the larger envelope reservoir is not being rapidly consumed. Its fuel content alone therefore does not determine the pulse's remaining duration.

The enriched envelope requires a hydrogen–helium atmosphere. A mixed atmosphere at $X = 0.9072$, $X_3 = 0.004$, $T_{\text{eff}} = 4600$ K and $\log g = 5.65$ satisfies independent flux, hydrostatic and chemical checks. At the stellar matching depth $\tau_{\text{Ross}} = 25.12$ it gives $T = 5440$ K and $P_{\text{gas}} = 3.136 \times 10^7$ dyn cm $^{-2}$. A surrounding hydrogen–helium-4 grid contains thirty solved columns covering $0.78 \leq X \leq 0.9955$, $4500 \leq T_{\text{eff}} \leq 5500$ K and $5.55 \leq \log g \leq 5.80$. Thirty-six further columns at $\log g = 5.40$, 5.25, 5.10 and 4.95, with temperatures 4700, 5000 and 5500 K, provide a local 54-column grid spanning $4.95 \leq \log g \leq 5.80$. At 5500 K, $X = 0.87$ and $\log g = 5.40$, increasing the achieved bottom optical depth from 876.3 to 1846 changes matching temperature by -0.002093% and gas pressure by -0.009951% . Two independent 4800 K columns at $\log g = 5.40$, with $X = 0.87$ and $X = 0.9955$, test interpolation between the 4700 and 5000 K sources. The mixture-to-hydrogen matching-state ratios differ by 0.09612% in temperature and 0.02421% in pressure. These are local source errors, not measured errors in the stellar trajectory. At $X = 0.8215$ and $X_3 = 0.007105$, an independent atmosphere with the actual isotopes agrees with the grid interpolation to 0.08784% in matching temperature and 0.03686% in pressure. A separately solved helium-4 mixture that preserves element number ratios, with the corresponding gravity adjustment, differs from the actual-isotope calculation by only 0.0003135% and 0.0007755%, respectively. Thus interpolation contributes more than the isotope approximation at this point. Five independent columns give largest differences of 0.08784% in temperature and 1.094% in pressure; these local comparisons do not bound the whole grid. At $X_3 = 0.0075$, $T_{\text{eff}} = 4500$ K and $\log g = 5.80$, a second isotope comparison gives differences of 0.00009978% in temperature and 0.0009838% in pressure. Lower-boundary comparisons at 5000 and 5500 K change the matching temperature by at most 0.02452% and gas pressure by at most 0.03430% at the stellar matching depth. These are local measurements. At $\log g = 5.25$, independent lower-boundary comparisons at 4700, 5000 and 5500 K change matching temperature by at most 0.004790% and gas pressure by 0.03703%. At $\log g = 5.10$, the corresponding limits are 0.003066% in temperature and 0.02347% in gas pressure. At $\log g = 4.95$, the corresponding limits are 0.005239% and 0.02060%. Nine sources at 6000 K span $X = 0.78$, 0.87 and 0.9955 at $\log g = 4.95$, 5.10 and 5.25. Together with the lower-temperature sources, they form a local 36-column grid covering 4700–6000 K and $4.95 \leq \log g \leq 5.25$. Independent depth comparisons at $\log g = 5.25$ change matching temperature by at most 0.01078% and gas pressure by 0.06911%. A 6000 K source with $X = 0.9955$ at the published hydrogen grid's lower boundary, $\log g = 5.50$, gives matching temperature 8341 K; a deeper calculation changes it by 0.003191% and gas pressure by -0.04062% . These source checks do not determine the absolute error of a hydrogen atmosphere inferred below the published gravity range. The late sequence uses a local grid of 36 solved atmosphere columns over $6000 \leq T_{\text{eff}} \leq 6500$ K and $4.20 \leq \log g \leq 4.95$, with the same three hydrogen fractions. Independent depth comparisons at $\log g = 4.20$ change matching temperature by at most 0.01095% and gas pressure by 0.09369%. At $\log g = 4.95$ and 6500 K, deeper source calculations change matching temperature

by at most 0.01373% and gas pressure by 0.09599%. The FreeEOS tables cover the corresponding lower densities. Source states, thermodynamic derivatives, pressure inversion and finite stellar steps are checked; existing overlapping table values are retained exactly. Sampled atmosphere derivatives and EOS pressure inversions pass their stated checks, including the higher-gravity cell entered during contraction. These establish local numerical consistency, not absolute atmosphere accuracy throughout the source rectangle. The absolute hydrogen reference below $\log g = 5.50$ remains inferred. From $\log g = 5.25$ to 5.10, its gravity dependence follows the calculated $X = 0.9955$ source response, anchored at the matching state at 5.25. A continuation of the fixed logarithmic slope supplies a comparison. Both prescriptions preserve every reference value at $\log g \geq 5.25$. A matched 10^4 yr calculation reaches $\log g = 5.231$, inside the lower-gravity interval. The prescriptions differ by 1.476 K in effective temperature and 0.1171% in surface luminosity; deposited nuclear energy differs by $7.675 \times 10^{-6}\%$, and the convective regions agree. This measures sensitivity along that interval, not an absolute atmosphere error. Above 5500 K, two continuations of the hydrogen reference preserve the matching values at 5500 K. One uses the calculated gravity response at 6000 K, anchored to the published hydrogen grid at $\log g = 5.50$; the other retains the 5500 K normalization and applies the calculated temperature response. Over a matched 8000 yr interval ending at 5553 K, they differ by 4.301 K in surface temperature, 0.3070% in surface luminosity and $5.576 \times 10^{-5}\%$ in deposited nuclear energy, with identical convective regions. This tests the part of the temperature interval occupied by the model, not the whole interval or the shared absolute uncertainty. A matched 12,000 yr comparison tests two hydrogen-reference continuations below $\log g = 4.95$, reaching 5701 K and $\log g = 4.882$. The effective temperatures differ by 0.2296 K and surface luminosities by 0.01569%; deposited nuclear energies differ by $9.917 \times 10^{-6}\%$, with the same final convective regions.

A second matched 10^4 yr calculation tests the inferred reference below $\log g = 5.10$ and reaches $\log g = 5.055$. The two prescriptions differ by 0.2991 K in effective temperature, 0.02206% in surface luminosity and $1.010 \times 10^{-5}\%$ in deposited nuclear energy, with the same convective regions. This measures the sensitivity to these two prescriptions over the sampled interval; it does not bound their shared physical uncertainty. For the inferred reference on $5.25 \leq \log g \leq 5.50$, a matched 1000 yr comparison of logarithmic continuation and calculated gravity response, starting at $\log g = 5.335$ and ending at $\log g = 5.331$, changes surface temperature by 11.27 K and luminosity by 0.9186%, while deposited nuclear energy differs by 0.02255%. The physical convective regions are identical. This tests the lower-gravity region occupied by the model, without establishing a uniform atmosphere uncertainty over the full extension. These stellar results use the differential helium response of this grid on the published hydrogen boundary. The difference between the absolute atmosphere prescriptions, and the pulse's spatial convergence, remain uncertainties; this extension is not selected for the main evolutionary track.

The surface abundances describe an isolated, nonrotating model without external accretion or magnetic fields. Accretion can maintain atmospheric metals against settling; the result depends on the supply rate, mixed-envelope mass and settling time [26]. The finite-convection pulse calculation sets semiconvective and thermohaline mixing coefficients to zero. The reaction ${}^3\text{He} + {}^3\text{He} \rightarrow {}^4\text{He} + 2p$ can lower the mean molecular weight and produce an inversion that drives thermohaline mixing, as found in red-giant models [27]. Whether this occurs strongly enough to alter the present shell requires an assessment of its composition gradient and transport times, rather than assuming that the highly stratified envelope persists.

There is also a difference between the atmosphere prescriptions themselves. Near $T_{\text{eff}} = 4559$ K and $\log g = 5.686$, at a helium mass fraction of 0.0045 and the same matching depth, the TLUSTY boundary gives a temperature 3.023% lower and a gas pressure 90.99% higher than the hydrogen boundary tabulated in MESA using the treatment of Rohrmann et al. [3, 2]. Joining their absolute values over a narrow composition interval would impose a substantial change of boundary prescription. A comparison instead applies the calculated mixed-to-hydrogen-rich temperature and pressure ratios to the

hydrogen boundary. This retains its normalization, but requires testing whether the composition response transfers between prescriptions. The atmosphere dependence of the pulse therefore remains unresolved.

A related physical example occurs in slowly accreting white dwarfs: helium-3 can accumulate through incomplete proton–proton burning and ignite a convective layer before the principal nova outburst [25]. The fuel supply and stellar structure differ from those of the isolated low-mass remnant, so this precedent supports the mechanism without establishing the present pulse’s amplitude or outcome.

Numerical treatment of the moving convective boundaries also matters. The cited MESA method holds the chemical mixing coefficient at its beginning-of-step value during each implicit solve [28]. An Ember comparison applies this treatment to finite convective and gradual chemical mixing while retaining the composition enthalpy flux. One interval of 1.6 yr and its subdivisions into two, four and eight steps pass the species and energy checks. Successive changes in integrated nuclear energy are 0.09470%, 0.04722% and 0.02357%. Pointwise composition differences remain within the imposed tolerance but do not decrease in this comparison; convergence of the complete pulse is not established by this local test.

Gradual mixing in layers stabilized by composition requires a separate physical assessment. Mixing-only controls with efficiencies of 0.01 and 0.1 retain a radiative thermal gradient in those layers. At their most sensitive active convective interfaces, the composition contribution to buoyancy is 3.503% and 31.68% of the parcel’s squared buoyancy contrast, compared with 0.3667% in the finest-grid model at 6803 yr. These ratios are diagnostics, not measured errors. They identify where the conventional composition-shifted mixing-length formula and separately transported composition enthalpy require a consistent treatment of parcel heat loss. The uncertain gradual-mixing controls therefore do not establish a preferred pulse trajectory.

The gradual-mixing calculation with efficiency 0.01 reaches 3004 yr, with nuclear power 1.352×10^{35} erg s^{−1} and maximum temperature 15.13 MK. Its bulk response through a convective-region merger is followed with an explicit absolute helium-4 error allowance of 2×10^{-6} ; the hydrogen, helium-3, structure and conservation limits are retained, with an additional 0.1% limit on endpoint nuclear-power differences. Full equation-of-state evaluations preserve the adjacent convection criteria, and a matched one-year propagation does not amplify the remaining trace-helium-4 difference. The merger’s timing and energy are less secure: four and eight substeps over its 0.5 yr interval differ by 10.17% in integrated nuclear energy. This uncertainty prevents using the calculation to infer a precise flash peak or completed energy release.

The burning shell is only partly degenerate. Before shell convection, the cell contributing most nuclear power lies at $m/M = 0.9612$ and 10.72 MK. Its thermal expansion coefficient $\delta = -(\partial \ln \rho / \partial \ln T)_P = 0.9099$ is close to the ideal-gas value. At fixed pressure and composition, the logarithmic nuclear heating response to temperature is 16.52. The central 90% of the nuclear power occupies a layer with thickness 0.05541 times its outer radius. Strong temperature sensitivity and thin-shell geometry therefore support a thermal instability without requiring strongly degenerate shell material. Heat transport, expansion and composition changes must be included to establish its growth and saturation.

Integrating the accepted nuclear powers through 3422 yr gives 7.964×10^{45} erg on the 666-point grid; through 3334 yr the 848-point grid gives 7.645×10^{45} erg. These integrals end at different ages and describe the computed portion of the pulse, not its completed energy release or a resolution error bar. Nuclear energy plus escaping neutrinos agrees with the change in nuclear rest mass. About 5% of the initial helium-3 inventory has been consumed; burning that entire inventory through helium-3 plus helium-3 would yield 1.561×10^{47} erg. This fuel scale does not predict how much will participate. Surface radiation carries about 7×10^{41} erg over the computed interval, so the nuclear release chiefly changes the structure.

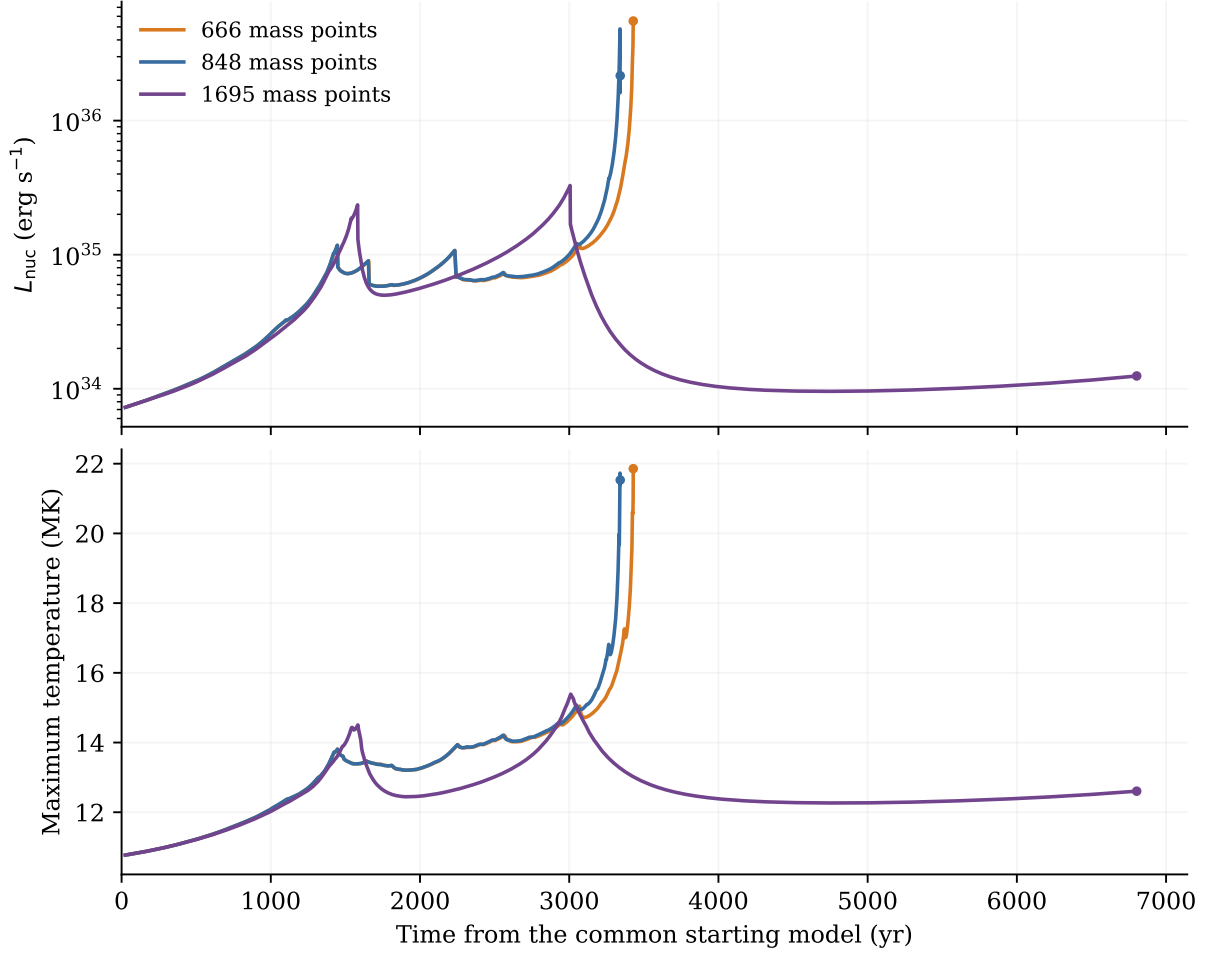


Figure 6: Nuclear luminosity and maximum internal temperature with total energy and composition enthalpy transported across the same mass surfaces. The three grids start from the same underlying stellar profile; adding points requires a small mechanical readjustment. All use the supplied white-dwarf atmosphere and finite convective mixing, with no semiconvective mixing. Time is measured from the common profile before shell convection. The 666- and 848-point endpoints reach 21.86 and 21.53 MK, respectively. The finest grid reaches 12.60 MK after a long decline and modest reheating. Sharp decreases near 3006 and 3341 yr accompany convective-region mergers. Different endpoints reflect the intervals computed so far. The full pulse amplitude, subsequent cooling and atmosphere sensitivity remain undetermined.

Figure 7 follows the internal response through 3422 yr. At its endpoint, the central 90% of nuclear power comes from $m/M = 0.9515\text{--}0.9573$, a layer containing 6.046×10^{-4} solar masses. Its radius spans $1.872 \times 10^4\text{--}2.035 \times 10^4$ km, its temperature 16.49–19.95 MK and its density $132.8\text{--}181.5$ g cm $^{-3}$. The layer’s mass-averaged fractions are $X = 0.1818$, $X_3 = 0.001793$ and $X_4 = 0.8164$, with negligible metals. Most remaining helium-3 lies outside this actively burning layer. The stellar radius grows from 4.403×10^4 to 5.003×10^4 km, while the center remains near 6 MK. At the strongest local burning, the instantaneous helium-3 consumption time is about 20 yr if temperature, density and composition are held fixed. Mixing and the changing thermal structure prevent this local estimate from specifying the duration of the complete event.

Species conservation and full-step versus two-half-step comparisons are checked throughout. During the pulse, the independent first-law residual is measured against the larger of total nuclear power and

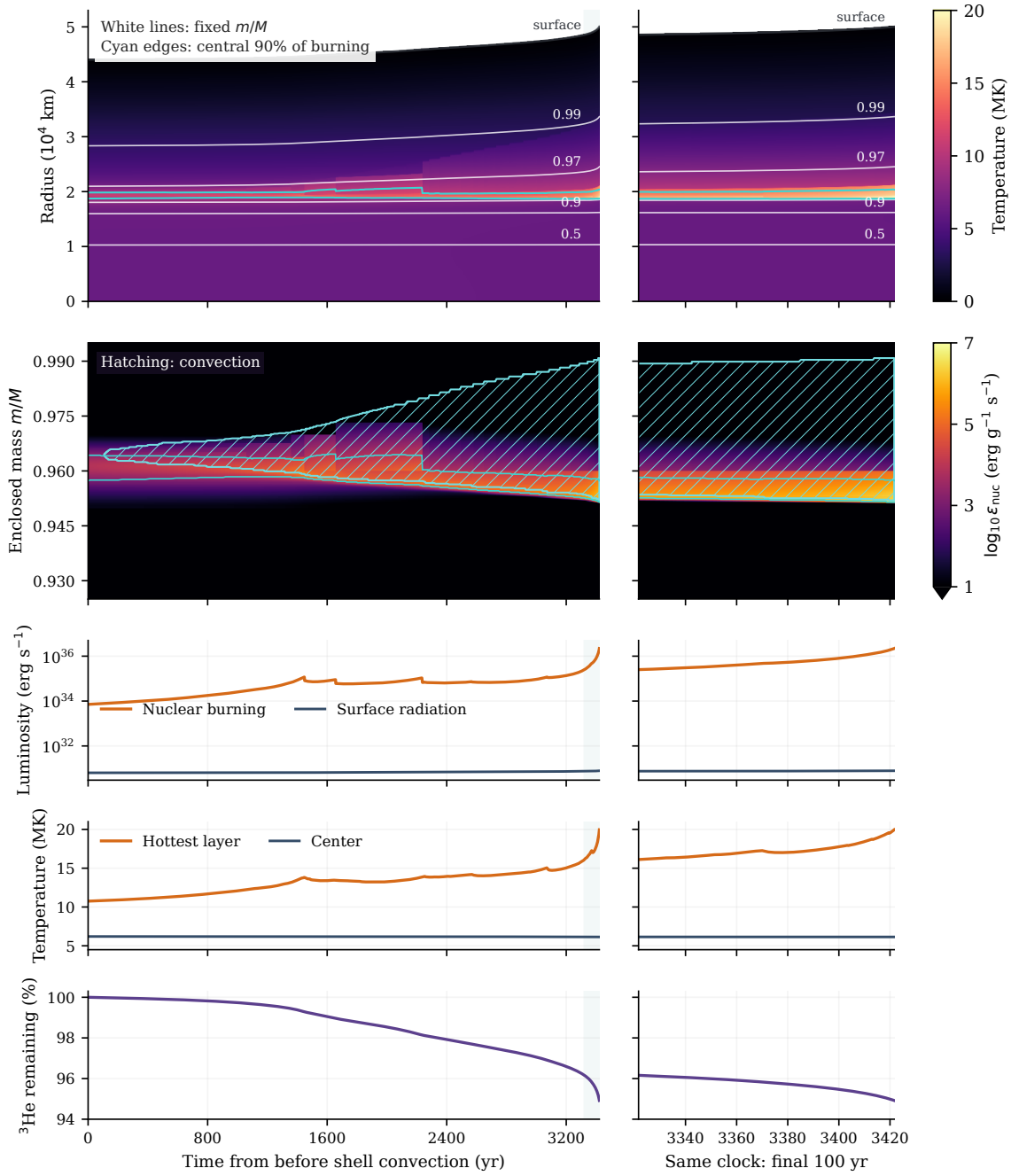


Figure 7: Internal development of the helium-3 pulse on the 666-point grid. The left column spans the computed interval; the right enlarges its final 100 yr on the same clock. Upper panel: temperature versus radius, with white curves following fixed enclosed mass fractions and cyan curves bounding the central 90% of nuclear power. Second panel: specific nuclear heating versus enclosed mass, with connected convective regions hatched. Lower panels show luminosities, temperatures and remaining helium-3. Time zero is an already-burning model before shell convection. The sequence ends during a renewed rise; the final peak and shutdown remain undetermined. The spatial sensitivity is shown in Figure 6.

photon luminosity, with a relative limit of 2×10^{-8} ; the photon-relative residual is also recorded. This avoids demanding excessive cancellation accuracy when nuclear heating greatly exceeds the escaping luminosity. The atmosphere comparison remains separate from the stated main trajectory.

This comparison also uses a bounded dense-core opacity extension over 0.25–0.45 keV. Radiation carries at most 0.2535% of the local temperature-gradient transport in the affected sampled layers; conduction dominates. Matched 1 Myr calculations with the nominal, unsuppressed and more strongly suppressed opacity give indistinguishable surface temperatures at the quoted precision. This local check supports continuation through the interpolation boundary, rather than a general opacity-accuracy bound.

In the dense hydrogen envelope, the OPLIB table used for the hydrogen dependence of opacity ends at $\log R = 1.5$, where $R = \rho/(T/10^6 \text{ K})^3$ in cgs units. A limited extension uses the hydrogen slope of the retained TOPS source, blending into it over $\log R = 1.25$ – 1.45 . It is restricted to $\log T = 5.6$ – 5.85 and $\log R \leq 1.8$. The temperature coverage continues through $\log T = 6.1$ using the same bounded hydrogen response; independent derivative and 50 Myr stellar comparisons support this extension. Differences on sampled common source coverage are at most 3.324%. Near the affected layer at 399 kK, convection carries 99.96% of the heat. A matched 50 Myr stellar calculation changes surface temperature by less than 10^{-5} K. This supports the local approximation without establishing a general opacity error bound.

Metals accumulate in the conductive core, reaching $Z = 0.1312$ at the plotted endpoint. Additional FreeEOS sources extend composition coverage to $Z = 0.16$. Direct source checks across a complete 512-point stellar profile and 18 additional core conditions have a maximum material-state difference of 0.07726%. A matched 25 Myr comparison changes effective temperature by 0.0004595 K and luminosity by 0.0002080%. In a model on the track, radiation supplies at most 0.04154% of gradient-driven heat in the region with $Z > 0.06$. Independent OPLIB data through $Z = 0.2$ support linear radiative-opacity scaling to within 3.302% over sampled 6–10 MK states within their density coverage. These data do not cover the densest core. Halving or doubling the core radiative opacity has no significant stellar effect over the tested 25 Myr interval; this supports bounded continuation with the stated opacity approximation, without bounding the accumulated cooling-age uncertainty.

The warm and intermediate-temperature interior opacities continue $\log \kappa$ in atomic hydrogen abundance from actual $X = 0.70$ and 0.75 sources through the pure-hydrogen limit. Below $Z = 0.01$, atomic opacity is continued linearly from the $Z = 0.01$ and 0.02 sources at fixed $X/(1 - Z)$, through zero metals. The hotter source retains its OPLIB hydrogen-composition response. Actual hydrogen-rich endpoint tables from $\text{\AA}SOPUS$ and OPLIB supply the added composition coverage. The molecular source ends at $1 - X = 10^{-7}$; a continuation across this remaining interval changes sampled opacity by at most 1.944×10^{-7} fractionally. The source values and physical stellar abundances are retained. Original temperature and density bounds remain enforced. A withheld warm-source composition comparison over a hydrogen interval of 0.195 differs by at most 11.01%; four common-domain OPLIB comparisons differ by 31.52–35.21%. Perturbing warm opacity by $\pm 15\%$ over 1.250 Gyr or by $\pm 40\%$ over 10 Myr changes luminosity and radius by less than 10^{-10} fractionally, with the sampled layers convective. At 3.695 Tyr, a separate $\pm 10\%$ radiative-opacity perturbation restricted to the depleted envelope changes luminosity by at most 0.6315%, radius by 0.1664%, and effective temperature by 3.216 K over 1 Myr, with the convective boundary moving at most one mesh point. Both perturbed evolutions pass their time and conservation checks. These are local stellar sensitivities, not universal opacity-error bounds.

Every accepted evolutionary interval compares one full step with two half steps. Species balances are checked independently in each composition region, together with total inert-metal mass, nuclear rest-mass release and the first law. The final convective core contains less than 10^{-9} of the stellar mass before becoming radiative. A 1 Myr interval across its disappearance passes the same pointwise composition and energy checks; the changing number of convective regions does not remove those tests.

The relative time-error budgets are 0.0003 for radius, density and temperature, 0.003 for local helium-3 abundance, and 0.001 for the other species. A matched 5 Gyr comparison against threefold smaller structure and helium-3 budgets uses 70.78% less solver CPU, with luminosity differing by 0.002872%, surface temperature by 0.01705 K, and total remaining hydrogen by 0.0001703%. The largest local species difference is about 0.2%. This comparison measures accumulated numerical differences over that interval, not a full-track error. Independent mass and energy checks retain their original thresholds. The nonlinear abundance solver uses a 10^{-10} tolerance with a stricter fallback; a matched 10^{-12} control changes mass fractions by at most 1.360×10^{-10} . Four reusable evaluation threads reproduce the two-thread physical results. Specializing the interpolation routine for its three derivative counts reduces solver CPU time by 8.194% over 1.5 Gyr, with every accepted structure and age unchanged. When a step fails to converge within the coupling-iteration limit, the controller reduces its size directly. Retrying such steps at a tighter tolerance is omitted; a stricter retry remains available for failed conservation checks. Over a matched 700 Myr interval, this change reduces solver CPU time by 21.03%, again with identical accepted structures and ages. These reductions are measured separately and should not be added.

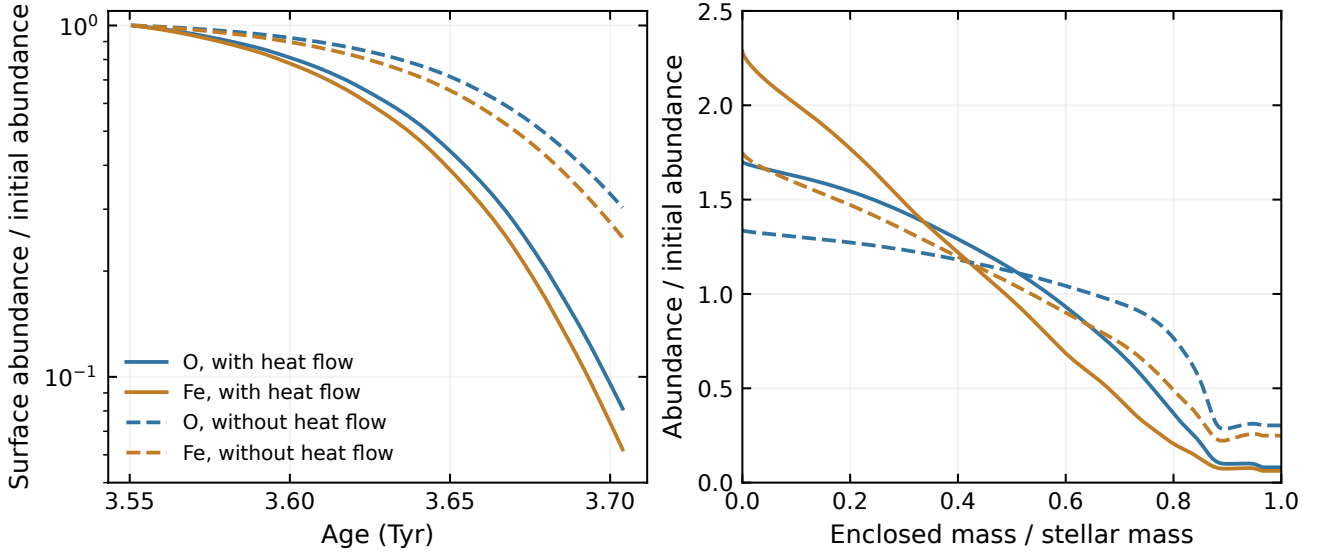


Figure 8: Passive oxygen and iron redistribution on prescribed stellar structures. Left: surface abundance relative to its initial value. Right: the abundance profile at 3.704Tyr. Solid curves include classical heat flow; dashed curves suppress it. Each tracer’s total inventory is conserved, and the changing convective envelope is mixed by mass. The two heat-flow treatments do not define an uncertainty band, and neither includes the star’s response to the changing metal opacity.

1.4 Comparison of the starting model with observed stars

The starting radius, luminosity and temperature can be compared with nearby low-mass stars and an eclipsing companion very close to $0.1M_{\odot}$. The following values are reported by Davis et al. [10], Agol et al. [11], and Ribas et al. [12].

Star	Mass ($M_{\odot}$)	Radius ($R_{\odot}$)	T_{eff} (K)	Luminosity ($L_{\odot}$)
Ember, initial	0.1	0.1259	2766	0.0008362
EBLM J2114–39 B	0.0993 ± 0.0033	0.1250 ± 0.0016	—	—
TRAPPIST-1	0.0898 ± 0.0023	0.1192 ± 0.0013	2566 ± 26	0.000553 ± 0.000019
Proxima	0.120 ± 0.003	0.146 ± 0.007	2980 ± 80	0.00151 ± 0.00008

The strongest comparison is EBLM J2114–39 B: its mass agrees with $0.1M_{\odot}$, and Ember’s radius is only 0.8% larger, within the reported radius uncertainty. The binary’s nearly solar metallicity, $[\text{Fe}/\text{H}] = -0.001 \pm 0.032$, also makes it relevant to our adopted composition. Its mass and radius are inferred from eclipses and orbital velocities together with an estimate of the primary’s radius; its temperature and luminosity are not measured in that study.

Ember falls between the lower-mass TRAPPIST-1 and higher-mass Proxima in radius, temperature and luminosity. These comparisons are less direct: TRAPPIST-1’s adopted mass uses an empirical mass–luminosity relation, and the quoted Proxima mass comes from stellar models. Age, composition and magnetic activity also differ between individual stars. Within Ember, roughly ten billion years of hydrogen burning changes luminosity by less than 0.4% and temperature by less than 2 K, so ordinary main-sequence age does not materially change this comparison. Formation and early contraction are excluded. Together, the radius agreement at nearly the same mass and the plausible temperature and luminosity support using this initial stellar structure for the calculation. They do not yet establish percent-level accuracy or select uniquely among the atmospheric and interior physics assumptions.

2 Independent calculations with MESA

We evolve the same initial mass and bulk composition with the unmodified MESA release 26.4.1, using its supplied physics and ordinary input controls [1]. The models begin with pre-main-sequence contraction. The grey-atmosphere calculation reaches a nuclear luminosity equal to 99% of the surface luminosity at 951.3 Myr, with $T_{\text{eff}} = 3076$ K. Its subsequent evolution is calculated with and without microscopic element diffusion, from the same saved structure. Both use the supplied pp and CNO reaction network, GS98 composition, OPAL and low-temperature opacities, electron conduction and an Eddington grey atmosphere integrated with the local opacity. The mixing-length parameter is 2.

The calculation without diffusion crosses central $X = 0.001$ near 2.372 Tyr. Its surface temperature reaches 6766 K at 2.406 Tyr, then decreases to 595.9 K by 2.564 Tyr. At the final state the radius is $0.02572R_{\odot}$, luminosity is $7.516 \times 10^{-8}L_{\odot}$, surface hydrogen is $X = 0.08778$ and the total remaining hydrogen mass is $0.0009172M_{\odot}$. The cooling branch uses the supplied time-centered gravitational-energy form of the energy equation. This is a computed cooling branch under the grey boundary prescription; it does not establish that prescription’s accuracy for a cold, helium-rich atmosphere. Evolution ends when the atmosphere requests local temperatures below the Ferguson opacity table’s 501.2 K lower limit. The Eddington relation predicts a corresponding surface temperature near 596 K, consistent with the calculated endpoint. This limit applies to the stated configuration.

With microscopic diffusion, the grey MESA sequence reaches 1.225Tyr, $T_{\text{eff}} = 3471$ K and central $X = 0.4593$. It uses 29.91 CPU-hours including 4012 retries. The repeated failures occur in the atmospheric equation-of-state inversion. Diagnostic calls request unphysical trial gas pressures, with $\log_{10}[P_{\text{gas}}/(\text{dyn cm}^{-2})] = 32.79\text{--}45.13$ at about 2919 K. Thus these failures do not establish an equation-of-state coverage gap along the accepted stellar structure. Holding the timestep fixed longer after a retry reduces retries in a matched 500 Myr calculation from 16 to 9, but reduces total CPU use by only 0.9422%; it does not resolve the slowdown.

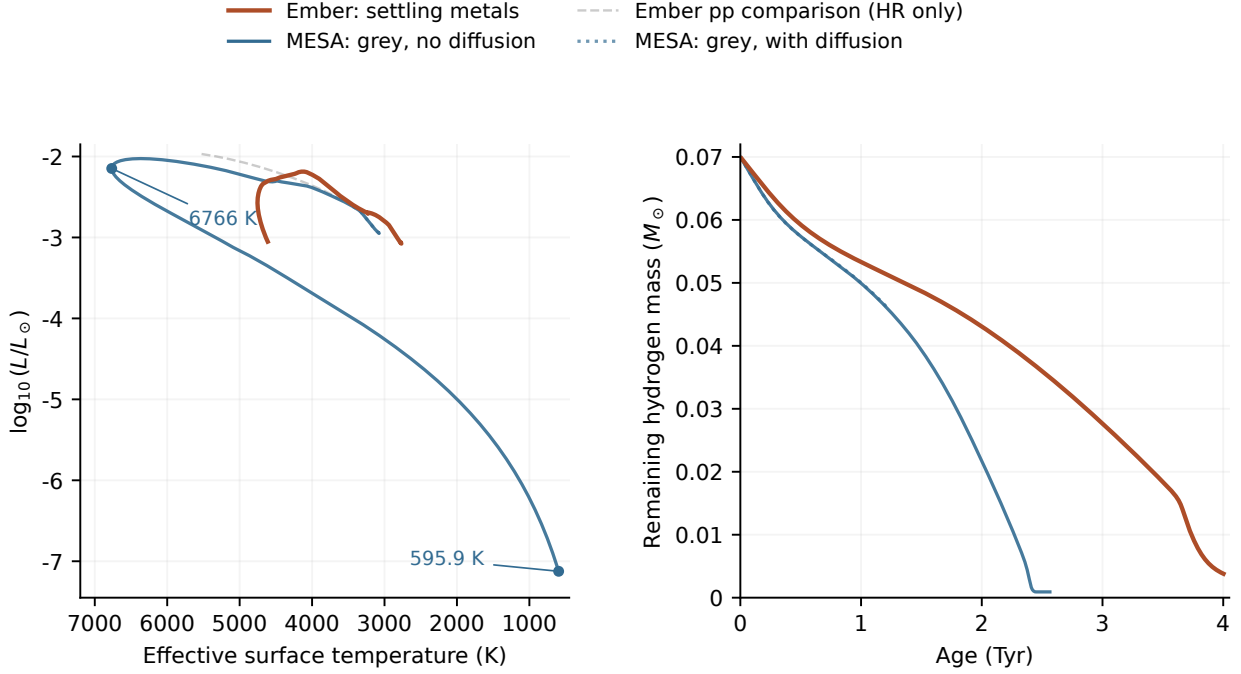


Figure 9: Ember and independent MESA calculations at $0.1M_{\odot}$. The left panel includes the computed grey MESA cooling branch; the right shows the remaining total hydrogen mass. Solid blue denotes MESA without diffusion; dotted blue denotes the same grey calculation with diffusion, through 1.225 Tyr. Orange is the same Ember trajectory as Figure 1. The faint dashed grey HR curve is Ember’s pp comparison without microscopic separation. The MESA curves shown here use grey atmospheres, not the separate tabulated-atmosphere configuration. MESA ages include contraction, whereas Ember ages start at its static hydrogen-burning model. The MESA contraction phase preceding the common starting model is omitted from these curves.

An Ember calculation with the same Eddington grey boundary prescription and no microscopic diffusion is shown in Figure 10. It retains convection and pp+CN burning, with Ember’s own equation of state and opacity sources. The initial homogeneous model has $T_{\text{eff}} = 2844$ K. The computed sequence reaches 3.742Tyr, at $T_{\text{eff}} = 2252$ K and central $X = 7.991 \times 10^{-7}$. Its core transports heat mainly by electron conduction; its envelope remains convective. The central temperature is 1.971 MK. The dense-core equation-of-state coverage extends through the current cooling model. Direct source comparisons support the sampled high-density region for $T \geq 1.8$ MK and $\rho \leq 4.96 \times 10^4 \text{ g cm}^{-3}$. Colder conditions require further assessment. The bend near 3241 K marks the formation of a radiative layer between the convective core and envelope at 3.305Tyr. Central helium-3 then declines as replenishment from the envelope ceases, accompanied by a luminosity dip. MESA shows the same type of transition near 4498 K and 2.295Tyr. Central hydrogen at onset is $X = 0.1728$ in Ember and $X = 0.08286$ in MESA. The first saved transition intervals span 3.346 Myr and 3.386 Gyr, respectively. The material physics and time resolution differ; the precise transient shapes have not been compared at matched accuracy, and the difference in transition temperature and composition remains unexplained.

In dense core cells lacking spectral opacity coverage, the radiative opacity uses the tabulated uncut value multiplied by an inferred plasma correction. The added nodes cover 0.25–0.45 keV, densities 10^4 – 10^5 g cm^{-3} , and $X \leq 0.2$. Electron conduction dominates in this region. A 500 Myr sensitivity test spanning uncut opacity through strongly suppressed radiation changes the endpoint by at most

0.0008222 K and 0.0001022% in luminosity. In the 2820 K model, the uncut-opacity limit bounds the local radiative share by 0.01249% in the sampled dense cells. These checks support the local approximation; they do not bound the full cooling-age uncertainty. For cooler core conditions, the same construction covers 0.125–0.225 keV, 10^4 – 10^5 g cm $^{-3}$ and $X \leq 0.2$. A matched 100 Myr comparison between uncut opacity and strong radiative suppression changes effective temperature by less than 10^{-7} K; electron conduction dominates these cells. Below about 3471 K, the grey calculation includes the quantum correction to nuclear screening described by Potekhin and Chabrier [40], using the Coulomb free-energy fit of Chugunov and DeWitt [41]. We restrict its use to screening parameter $\zeta \leq 0.5$ and one-component coupling parameter $\Gamma \leq 200$; this does not describe zero-temperature burning. A matched 100 Myr calculation changes surface temperature by 0.08920 K and luminosity by 0.01209%. The hotter portion retains classical screening; the local comparison does not bound its accumulated effect on age. The cooling calculation uses ten hydrogen fractions over $0 \leq X \leq 0.225$ and four helium-3 fractions, with finer temperature and density spacing where ionization changes rapidly. At sampled envelope states, the largest heat-capacity and adiabatic-gradient differences from direct sources are 1.588% and 0.5182%. At sampled inner-core states down to 1.8 MK, thermal coefficients agree within 0.5%, while pressure and internal energy agree within 10^{-6} . This inner-core bound does not apply to the whole star: a broader comparison at the current profile finds heat-capacity and adiabatic-gradient differences of 2.077% and 4.166% near $m/M = 0.7164$, at 1.900 MK. These measure interpolation errors in the sampled domain; their effect on cooling requires a finite stellar comparison. The calculation through this endpoint uses 515.1 solver CPU-minutes, summed over continuation segments and threads, including rejected and terminal stellar attempts but excluding process startup, table preparation and separate controls. A paired 500 Myr calculation tests a less restrictive internal composition iteration while retaining the same time-error and conservation requirements. CPU use falls by 27.54%, with endpoint differences of 0.001296 K and 0.0001392% in luminosity. This is a local performance measurement. Failed Newton or composition iterations trigger a smaller timestep before further boundary searches. The next proposal also retains the last successful interval. A 500 Myr comparison with 10 Myr reference steps changes the endpoint by 0.009721 K and 0.001110% in luminosity, with all time, species and energy requirements unchanged. The grey continuation uses the same independently sourced dense hot-opacity extension as the metal-settling calculation. A 25 Myr overlap calculation has identical physical outputs; an 85.63 Myr step into the extended domain passes the unchanged time and conservation checks. Trials outside the source domain are rejected before repeated boundary or tolerance searches.

At a remaining hydrogen mass of $0.04963M_{\odot}$, Ember reaches 1.302Tyr and MESA 1.019Tyr, with effective temperatures 3045 and 3403 K and luminosities $0.001626L_{\odot}$ and $0.002316L_{\odot}$, respectively. Their remaining helium-3 masses are $0.005740M_{\odot}$ and $0.003710M_{\odot}$: matching hydrogen alone does not match the nuclear energy already released. Ember uses 7.789 CPU-minutes through this point, compared with 4.912 CPU-minutes for MESA, a ratio of 1.586. Ember’s counter measures native stellar-solver work; MESA’s measures the full native process, including loading and output. Separate table construction, validation calculations, compilation and the MESA pre-main-sequence calculation are excluded. These fully convective intervals do not measure relative performance during later shell burning or white-dwarf cooling.

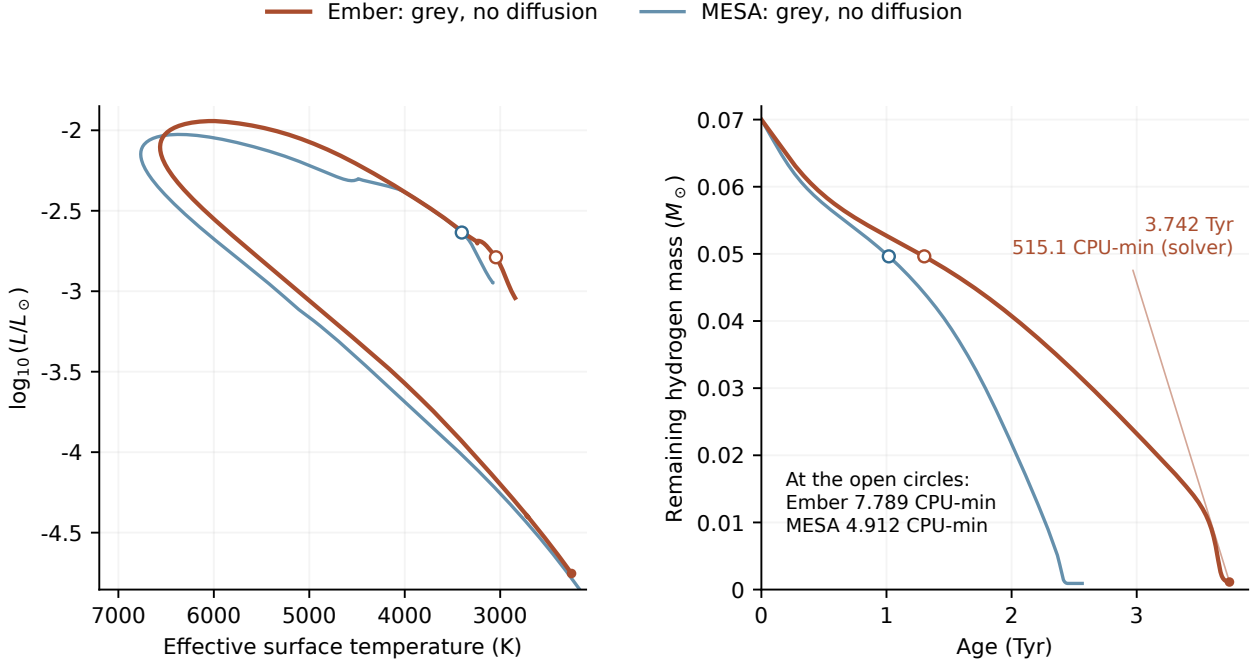


Figure 10: Grey-atmosphere calculations without microscopic diffusion. Both have initial mass $0.1M_{\odot}$, $X = 0.7$ and $Z = 0.02$. Orange is Ember through 3.742Tyr; blue is MESA. Open circles mark the same remaining hydrogen mass, $0.04963M_{\odot}$, used for the CPU comparison. The filled orange point marks the last computed Ember model in this figure; its label gives the cumulative 515.1 solver CPU-minutes for that sequence. The open-circle times refer only to the earlier equal-hydrogen comparison. The equations of state, opacities and reaction networks differ. CPU times have the scope defined in the text and are summed over evaluation threads. MESA’s age includes contraction; Ember starts from a static hydrogen-burning model. The small Ember luminosity dip near 3241 K accompanies the loss of full convection. The MESA cooling branch continues beyond the displayed H–R limits.

The MESA configuration with a broader set of supplied physics uses PHOENIX/Castelli–Kurucz atmosphere tables, OPLIB and Ferguson opacity, the FreeEOS/OPAL–SCVH/HELM/Skye equation-of-state blend, and conduction with the Blouin hydrogen/helium correction. It follows 34 nuclear species and individual-isotope diffusion with calculated ionization and diffusion heating. Convection uses the Ledoux criterion, Henyey mixing-length transport with parameter 1.9, and convective premixing; thermohaline mixing uses the Brown–Garaud–Stellmach prescription. The supplied carbon/oxygen phase-separation option is not applicable to this helium core. The initial hydrogen, helium and metal fractions remain 0.7, 0.28 and 0.02; the initial deuterium and helium-3 fractions are zero.

The tabulated-atmosphere calculation has reached 362 Myr of contraction, at 2986 K, with central $X = 0.7$. A supplied time-step control rejects changes in $\log_{10} T_{\text{eff}}$ above 0.02 per step, and diffusion uses at least two substeps per stellar step. All material and transport prescriptions remain as specified above. The supplied grey fallback is permitted for temporary solver trials; all 3430 saved models in the continuation from 263.9 to 362 Myr use the tabulated atmosphere without that fallback. Controlled calculations identify microscopic diffusion interacting with surface composition as a source of numerical difficulty in the configuration with a tabulated atmosphere. A deeper diffusion boundary through 10^5 K gives a fresh 734-model calculation to 10 Gyr without retries. Its continuation reaches 2.976Tyr, with central $X = 0.1512$, where diffusion-enabled trials produce a sudden temperature change. The

outermost radiative skin retains a different composition from the convective envelope; the diffusion routine homogenizes it when the interior diffusion domain becomes active. This changes the atmospheric boundary abruptly. A stock mixing coefficient restricted to the outer 10^{-6} of the mass removes the interruption in two 60-model tests, which reach 2.904Tyr with active diffusion and no retries. Its continuation reaches 2.986Tyr and 4694 K, with central $X = 7.123 \times 10^{-6}$, surface $X = 0.9903$ and active diffusion, in 21.72 additional CPU-minutes. Restricting the imposed mixing to the outer 10^{-7} rather than 10^{-6} of the mass gives identical results in a matched 60-model comparison. These tests support the numerical diagnosis, but do not establish a unique physical surface-mixing prescription or validate the inherited history. They are not substituted into the plotted MESA segment. A further diagnostic switches a hydrogen-rich model at 3.163Tyr, 4737 K and $\log g = 6.010$ from the gray boundary to the supplied white-dwarf table. A 1000 yr adjustment lowers photon luminosity by 19.96% and effective temperature by 5.410%, with negligible initial change in radius or nuclear power. With the ordinary temperature guard restored, the next 33.65 Myr reach 4176 K and a radius 15.33% larger than before the switch. This is an evolving thermal response, not an equilibrium uncertainty estimate. Below $\log g = 6$, MESA’s supplied wrapper blends the table with a gray boundary; the endpoint has 22.81% gray weight in its logarithmic boundary values. It is therefore not a pure white-dwarf-table continuation, and is kept separate from the plotted configurations.

A measured contraction calculation through 248.6 Myr uses 5.449 CPU-hours and 2.845 hours elapsed with two threads; overlapping restarts are not added to this cost. These contraction-stage costs are not a comparison at equal remaining fuel with Ember. The supplied stellar atmosphere tables also assume fixed hydrogen/helium mixtures and contain grey replacements for some missing cells. The supplied hydrogen white-dwarf table extends down to 2000 K and requires a suitable surface composition [2]. Successful table interpolation alone would not establish accuracy for an arbitrary hydrogen/helium mixture.

Computing times are measured on the same Apple M4 Max, using double precision. With the total-energy equation, the grey calculation from the common hydrogen-burning model through 750.6 K takes 1.727 CPU-hours, including retries and its stalled cold tail; constructing the common model takes another 0.4316 CPU-hours. With the time-centered gravitational-energy equation, the interval from before the temperature maximum through 595.9 K takes 0.4469 CPU-hours on two threads, excluding the shared preceding evolution. These overlapping measurements are not added together. The two energy forms agree in total age through 800 K to 0.0004883%; this numerical comparison does not measure uncertainty in the physical cooling age. Ember’s plotted metal-settling sequence uses 36.31 CPU-hours, including its independent time-step comparisons and necessary retries; its table construction is accounted for separately. The same retained Ember segments account for 15.13 hours of elapsed computation, excluding system sleep and pauses between continuations. This is a timing account of the computed track, not a complete cooling benchmark. These are different physical calculations and different evolutionary phases, so their ratio is not a measurement of relative solver efficiency. The complete pp/CNO network in the grey MESA calculation shows that including those reactions does not by itself require a large computation. Figure 10 compares equal remaining hydrogen with matched grey-atmosphere and diffusion choices, while retaining the distinct material and nuclear prescriptions.

The plotted pp and metal-settling Ember tracks change both nuclear burning and element transport, so their separation cannot isolate either effect. In a matched 50 Gyr Ember comparison, adding diffusion increases the endpoint luminosity by 20.10% and changes the effective temperature from 3302 to 3404 K, while the total remaining hydrogen differs by only 0.2767%. Diffusion therefore has a demonstrated structural effect after full convection ends. Together with the metal-settling comparison, this makes element redistribution a stronger explanation for the large late-track separation than an omitted oxygen branch alone. The oxygen branches still require inclusion and a separate comparison; a unique causal division of the full track difference has not been measured.

3 Comparison with the published LBA97 model

The main comparison is with Laughlin, Bodenheimer and Adams [13], here abbreviated LBA97. Their section 3.2 and Figure 1 describe the full hydrogen-burning evolution of a $0.1M_{\odot}$ star and its entry into white-dwarf cooling. We use numbers stated in the text and approximate values read from the published figure.

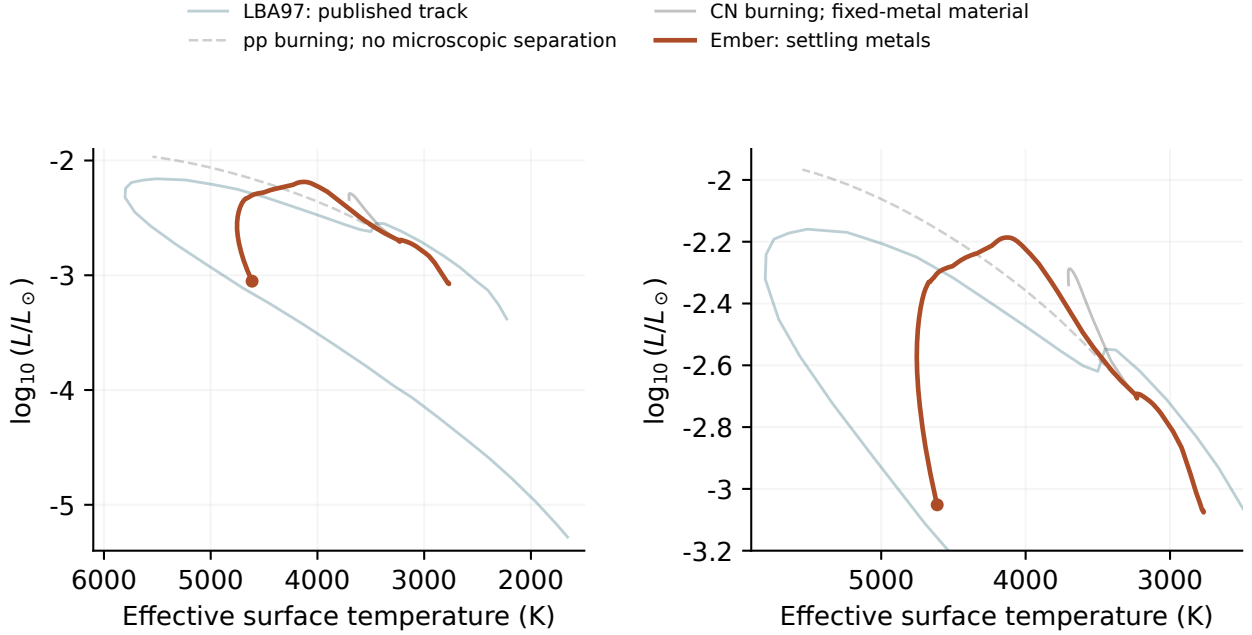


Figure 11: HR diagrams showing the calculation with metal settling through 4.002Tyr (orange), the fixed-metal CN/diffusion comparison through 3.704Tyr (solid gray), the pp comparison without microscopic separation through 3.890Tyr (dashed gray), and LBA97 (pale blue). The left panel shows the wider range and the right panel the hydrogen-burning portion. The orange curve uses the same accepted states as Figure 1.

Quantity	LBA97, published	Ember pp comparison
Initial surface temperature (K)	2228	2766
Initial luminosity ($L_{\odot}$)	$\simeq 0.000417$	0.0008362
Initial central density (g cm^{-3})	309	381.7
Maximum ${}^3\text{He}$ mass fraction	0.0995	0.1003
Age at maximum ${}^3\text{He}$ (Tyr)	1.38	0.7103
Nonconvective central core	Forms at 5.742Tyr	Present at 3.890Tyr
Hydrogen mass fraction at core onset	$\simeq 0.16$	Not yet measured
Highest surface temperature (K)	5807	Not yet reached

The two starting states are defined differently. LBA97 includes contraction and defines main-sequence arrival by nuclear reactions supplying all surface luminosity, after about 2 billion years. Ember starts from a static hydrogen-burning model at age zero. That clock offset is small compared with the later differences, but the initial thermal and atmospheric models also differ. Ember initially radiates about twice the published luminosity and has a hotter surface. These comparisons should not be interpreted as a measurement of the error in one isolated physical ingredient.

3.1 Luminosity and the nuclear timescale

An initially $0.1M_{\odot}$ star with $X = 0.7$ contains $0.07M_{\odot}$ of hydrogen. Taking about $6 \times 10^{18} \text{ erg g}^{-1}$ of deposited energy gives a nuclear budget of order $8 \times 10^{50} \text{ erg}$. Its burning time scales as this budget divided by its mean nuclear luminosity. During full convection, this can be written $dt \simeq -Mq_{\text{eff}} dX/L_{\text{nuc}}(X)$, with q_{eff} accounting for neutrino losses and energy temporarily stored in ${}^3\text{He}$. A persistently lower luminosity therefore lengthens the nuclear clock.

The initial Ember/LBA97 luminosity ratio is 2.006, close to the ratio 1.943 of their ages at maximum ${}^3\text{He}$. At their early luminosity minima, the F77 AJR and Ferguson models have $T_{\text{eff}} = 3218$ and 3278 K and luminosities 0.001331 and $0.001405L_{\odot}$. Their luminosity ratios relative to LBA97 are 3.193 and 3.370; the corresponding ${}^3\text{He}$ peak-age ratios are 3.323 and 3.632. The peak abundances differ, so this is an approximate comparison of stages. The ratios change later in evolution; a single time rescaling is insufficient.

A simple structural estimate gives the same scale. At fixed mass and composition, homologous ideal-gas interiors have $\rho_c \propto R^{-3}$ and $T_c \propto R^{-1}$. Taking $\epsilon_{pp} \propto \rho T^{\nu}$ with $\nu \simeq 4\text{--}6$ and equating nuclear power to $4\pi R^2 \sigma T_{\text{eff}}^4$ gives

$$R \propto T_{\text{eff}}^{-4/(\nu+5)}, \quad L \propto T_{\text{eff}}^{4(\nu+3)/(\nu+5)} \simeq T_{\text{eff}}^{3.1\text{--}3.3}.$$

The Ember/LBA97 temperature ratio then predicts a luminosity ratio 1.959–2.029, including the radius adjustment. Partial degeneracy and nonideal thermodynamics limit this estimate. The thermal times $GM^2/(RL)$ are only about 3.0 and 5.5 billion years, respectively; a starting thermal imbalance cannot itself sustain a trillion-year offset. The relevant difference is the equilibrium sequence maintained by the physics.

The low LBA97 luminosity was already a recognized issue in 1997. Their section 3.1 reports a poorer fit to the observed mass–luminosity relation than the Burrows models, while noting uncertainty in the bolometric corrections used for the Hyades comparison. The independent solar-metallicity model of Chabrier and Baraffe [14], at $0.1M_{\odot}$ and 1 billion years, gives 2811 K , $0.1240R_{\odot}$ and $0.0008511L_{\odot}$, close to Ember; its helium fraction is 0.275 rather than 0.28. Thus the cooler LBA97 model also differed from contemporary detailed atmosphere calculations. Chabrier and Baraffe find that a grey Eddington boundary generally makes low-mass models hotter, so the use of a grey atmosphere alone does not explain LBA97’s lower temperature. Controlled atmosphere, opacity and EOS substitutions are needed to identify its cause.

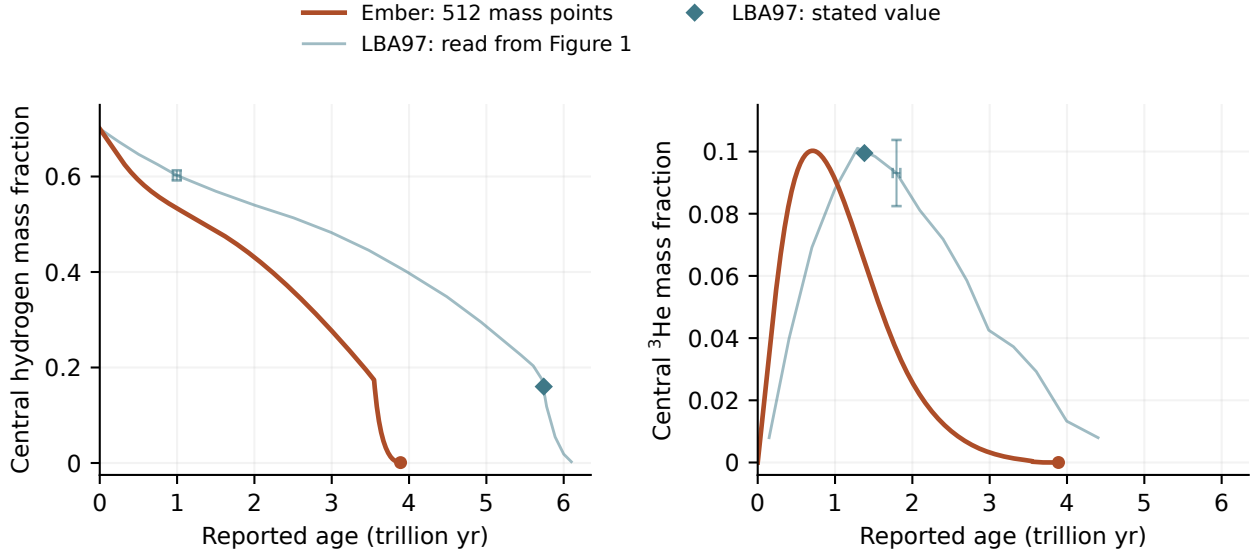


Figure 12: Hydrogen consumption and the buildup and depletion of ^3He in the pp comparison without microscopic separation. The pale curves are read from the inset of LBA97 Figure 1, with 20 hydrogen and 15 helium-3 samples. Diamonds show their stated ^3He maximum and the hydrogen abundance at central radiative-core formation. Orange curves are the 512-point Ember track through 3.890Tyr. Ages are shown as reported, without shifting or stretching either calculation. The representative crosses correspond to four image pixels: about 0.05Tyr in age and 0.01 in mass fraction. They illustrate reading precision, not statistical or physical uncertainty. The rapid final hydrogen decline is poorly resolved in the printed inset; the helium-3 trace ends as it becomes indistinguishable from the baseline.

The maximum ^3He abundance agrees to within about 1%, but Ember reaches it in about half the time. Ember also consumes hydrogen faster over the interval calculated so far. Reading the published hydrogen curve gives an age of roughly 5.6Tyr at hydrogen fraction 0.2034, reached by Ember near 3.40Tyr. The model at 3.890Tyr has central hydrogen below 0.16, the approximate abundance stated by LBA97 at its central-core transition. The figure does not justify a more precise age comparison. The luminosity–temperature track alone can conceal this difference because it does not show how long the star spends at each position.

LBA97 used a grey atmosphere, approximate opacity changes as hydrogen is consumed, and older interior opacities and nuclear rates. Ember uses frequency-dependent gas atmospheres, composition-specific opacity tables, a different equation of state and updated pp-chain rates. Grain formation is still omitted from Ember’s evolving atmosphere. These differences are candidates for the changed surface properties and hydrogen-consumption rate; comparisons that change one ingredient at a time are needed to separate them. Agreement with LBA97 and computing time alone do not establish physical accuracy.

3.2 Core transport by radiation and conduction

At the center of the metal-settling model at 3.688Tyr, electron conduction contributes 99.79% of the local transport driven by a temperature gradient. The conductive and radiative coefficients are 1.058×10^{14} and $2.221 \times 10^{11} \text{ erg cm}^{-1} \text{ s}^{-1} \text{ K}^{-1}$, respectively; heat carried with diffusing species is accounted for separately. The central electron degeneracy parameter is 15.39. These local values do not imply that radiation is unimportant in the envelope.

The pp comparison first departs from full convection near 3.546Tyr, at central hydrogen fraction $X = 0.1751$. By 3.560Tyr, the convective mass fraction has fallen to 0.6277 and central hydrogen to $X = 0.1518$, while surface hydrogen remains at $X = 0.1747$. This separation of central and surface composition shows that convection no longer mixes the whole star.

At 3.890Tyr, the model has a core containing 97.43% of the mass that transports heat by radiation and conduction, surrounded by a convective envelope. Central hydrogen has fallen to $X = 0.0008157$, while the envelope remains near $X = 0.1730$. Radiation and conduction can carry the required energy through this core without convection; the composition gradient also suppresses convection. Conduction supplies 91.29% of the local heat flux at the center, with radiation carrying the remaining 8.712%. Conduction is therefore already significant in the central region. Figure 13 shows the declining convective mass and the final abundance profile on this track. The first loss of full convection does not by itself establish a central radiative core. The final profile establishes the core but does not determine its formation age. LBA97 reports central radiative-core formation at 5.742Tyr.

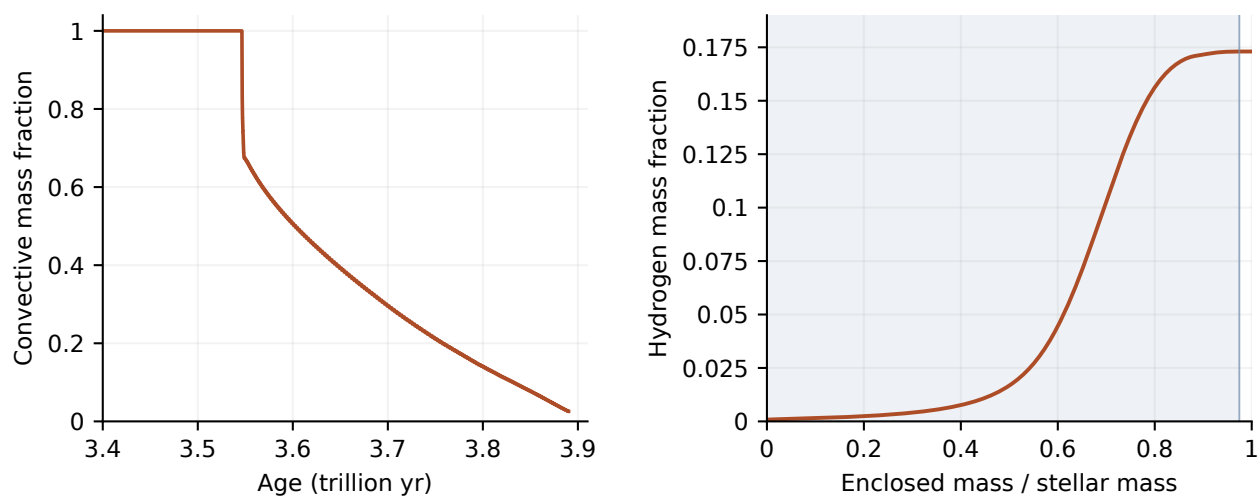


Figure 13: Convection and chemical separation by nuclear burning and macroscopic mixing in the pp comparison without microscopic separation. Left: the fraction of mass in convective regions declines from unity near 3.546Tyr to 0.02565 at 3.890Tyr. Right: the hydrogen profile in the 3.890Tyr model. The shaded core carries heat by radiation and conduction; the outer convective envelope has an almost uniform hydrogen abundance. Both panels use the same calculation and its final saved structure.

“Fully convective” describes where convection occurs, not the fraction of the luminosity it carries. Radiation, conduction and convection share the heat flow continuously. The model’s mixing-length calculation makes the convective contribution approach zero at a stability boundary. Chemical mixing uses a stronger approximation: each connected convective region becomes uniform rapidly. Slow composition mixing is included in some marginally stable regions, but their additional heat transport still needs attention.

3.3 The reconstructed F77 program is a separate comparison

The [Henyey/F77](#) program is useful for controlled code comparisons, but it does not reproduce the LBA97 calculation. Its AJR-opacity run first flags the center as nonconvective at 3.307Tyr and $X_{\text{H}} = 0.01756$; its Ferguson-opacity run does so at 2.946Tyr and $X_{\text{H}} = 1.064 \times 10^{-5}$. Both differ substantially from the published 5.742Tyr and $X_{\text{H}} \simeq 0.16$. The AJR run’s highest surface temperature, 5936 K, is much closer to LBA97’s 5807 K, but agreement in that one quantity does not validate the full history.

The F77 reconstruction substitutes some opacity and nuclear-rate prescriptions. At $X_{\text{H}} = 0.2034$, it is about 76% more luminous than Ember; switching only its low-temperature opacity choice changes its luminosity by 54.5%. These comparisons establish sensitivity to the inputs, rather than identifying the source of the Ember–LBA97 difference. The [separate comparison notes](#) and [supplementary F77 graphs](#) retain these results. The F77 curves are thin and translucent, with abrupt changes drawn more faintly and all accepted data preserved.

3.4 Photospheric evolution over the opacity table

Figure 14 follows the presentation of LBA97 Figure 6: photospheric density and temperature over an opacity map, with circle diameter proportional to stellar radius. For the pp comparison, we recover the photospheric state at Rosseland optical depth 2/3 from the accepted atmosphere structures. The local temperature there differs from the effective temperature of a frequency-dependent atmosphere. It rises from 3019 to 5708 K over the plotted history, while density falls overall from 1.432×10^{-5} to $2.558 \times 10^{-6} \text{ g cm}^{-3}$. The two opacity maps use the same axes and color scale. The LBA97 map is inferred from the published shading, calibrated against the tabulated opacities of Alexander, Johnson and Rypma [15]. Nineteen cells excluded from the calibration differ by at most 0.12 in $\log_{10} \kappa$; this checks the approximate scale of the diagram, not the accuracy of its grain physics.

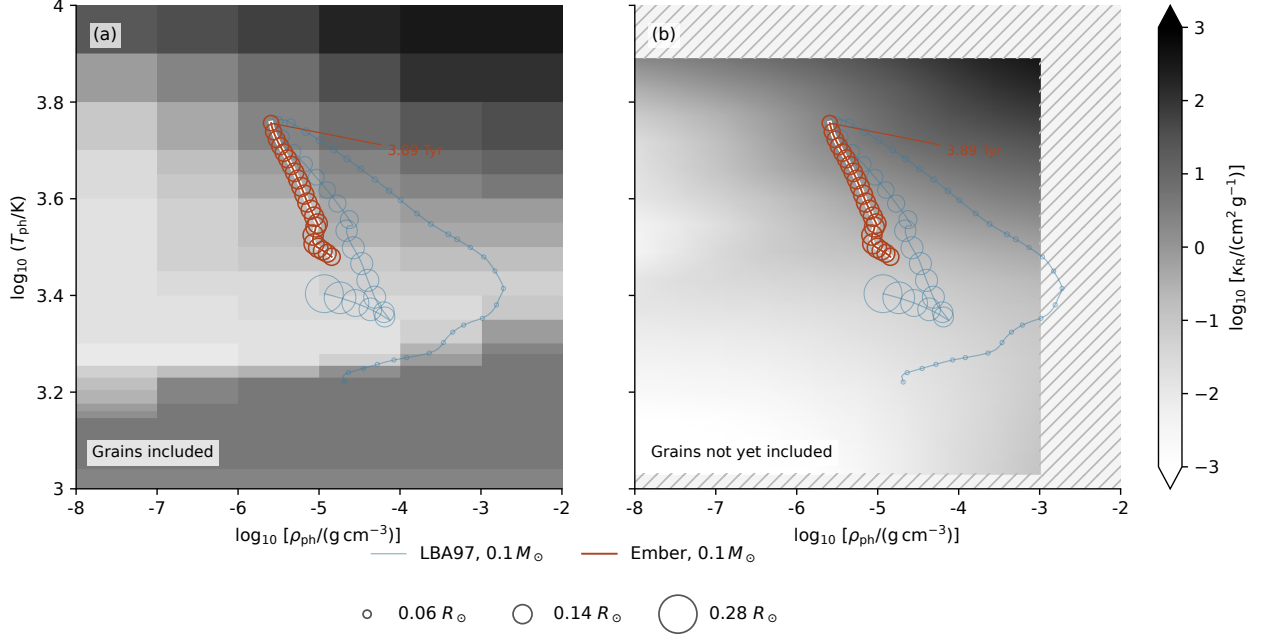


Figure 14: Photospheric evolution of the $0.1M_{\odot}$ pp comparison without microscopic separation, over hydrogen-rich opacity maps: (a) the grain-inclusive LBA97 Figure 6 map on an approximate calibrated opacity scale; (b) the Rosseland absorption mean of Ember’s SYNPEC wavelength table at $X = 0.7$, $Z = 0.02$ and zero ^3He . Both panels use identical axes and grayscale normalization. The right map excludes scattering and grains; the low-temperature grain-opacity band in the left map is therefore absent. Hatching marks missing wavelength-table coverage. Both panels show the LBA97 track recovered from 897 figure markers and Ember’s calculated track through 3.890Tyr. The LBA97 curve includes contraction and cooling; the plotted Ember models are burning hydrogen. Open-circle diameters scale with radius, with 48 LBA97 and 22 Ember samples. LBA97 radii are normalized using its stated main-sequence luminosity and temperature. Ember’s local photospheric temperature and density are interpolated at Rosseland optical depth 2/3 using the evolving surface composition. The fixed backgrounds do not represent the changing opacity of either star.

Photospheric temperature and density are interpolated in surface composition, effective temperature and gravity using the atmosphere’s molecular equation of state. The corresponding interpolation at the deeper stellar matching boundary reproduces its temperature and pressure to numerical precision at 65 sampled models.

4 Physical ingredients in Ember

Component	Treatment and present limits
Equation of state	FreeEOS 3.0 supplies pressure, internal energy and their responses to temperature, density and composition. Electron integrals are evaluated numerically. Interpolation is based on a shared Helmholtz free energy to keep these responses consistent. The selected 72 composition tables include zero hydrogen and the low-density states needed at the atmosphere boundary. They do not describe the cold, dense remnant.

Component	Treatment and present limits
Opacity	AESOPUS 2.1 provides low-temperature gas opacity, and the LANL TOPS service supplies ATOMIC opacities for hotter and denser conditions [8, 9]. The two are joined only where their available ranges permit it. Hydrogen-poor compositions have been refined and checked against additional calculations. The ATOMIC tables reach zero hydrogen; the cool opacity still requires hydrogen-rich mixtures.
Hydrogen burning	A reduced ppI/ppII network follows ${}^3\text{He}$ out of equilibrium. It uses Solar Fusion II reaction data and finite-degeneracy Salpeter–Van Horn screening, which accounts for the plasma’s effect on charged reactants. Atomic mass differences determine heat and nuclear-neutrino energy. Pep, hep, ppIII and CNO reactions are absent; metals do not react.
Heat and composition transport	Mixing-length convection uses the Ledoux criterion, including the stabilizing effect of composition gradients. Slow mixing associated with semiconvection and thermohaline instability is included. Conduction uses approximate mixtures of Ioffe/Cassisi inputs. Microscopic diffusion, settling and a separate semiconvective heat flux are absent.
Atmosphere	TLUSTY208/SYNSPEC54 calculates plane-parallel gas atmospheres in local thermodynamic equilibrium, resolving absorption and scattering over frequency. The interior joins the atmosphere at optical depth 100. The atmospheres include atomic and molecular absorption and collision-induced absorption, but no accepted condensate treatment.
Thermal neutrinos	Plasma-decay losses follow Haft, Raffelt and Weiss [19] and enter the energy equation with analytic derivatives. Other thermal-neutrino processes are not included in this choice.
Numerical evolution	Coupled implicit equations solve structure, burning and mixing. Comparing a full time step with two half steps controls time accuracy. The mass mesh is fixed. Saved models include enough internal information to continue with the same executable, tables and numerical settings.

The interior equation of state, atmosphere chemistry, opacity and conduction come from different physical sources. Their agreement therefore needs checking; accurate derivatives within one component do not prove that all components describe the same thermodynamics. FreeEOS omits potassium from the GS98 mixture, a missing mass fraction of 4.56×10^{-6} . Helium-isotope and atomic partition-function approximations also remain.

With all other inputs fixed, replacing the Solar Fusion II rates with Solar Fusion III [31] raises the initial luminosity by 0.7454% and radius by 0.2164%. Both rate choices have been evolved through 10^{10} yr for this comparison. The plotted trajectory uses Solar Fusion II; the short comparison does not determine a change in the full hydrogen-burning lifetime.

5 Why more atmosphere calculations are needed

The atmosphere calculations are reusable. They are performed separately and stored as tables of boundary temperature and pressure. During stellar evolution, Ember interpolates those tables; it does not solve frequency-dependent radiative transfer at every time step. New atmosphere calculations are needed when the star reaches a composition, surface temperature or gravity not adequately covered, or

when the physical treatment changes.

The plotted track uses 220 gas atmospheres, spanning the combinations of surface temperature, gravity and composition that it visits, with source models extending through 5600 K. The tabulated source states include hydrogen fractions from 0.1 to 0.7 and ^3He fractions from zero to 0.12. These outer ranges do not imply that every combination has been calculated: interpolation requires a complete set of surrounding source models, including those needed for derivatives.

Independent atmosphere calculations test the interpolation. At two 3700 K models with compositions between the tabulated values, matching temperatures differ by -0.4837% and -0.7399% , and gas pressures by $+0.5707\%$ and $+0.5137\%$. Changing the lower boundary of the source atmospheres changes matching values by at most 0.007883% in the specified comparisons. All 220 source models have support in the selected 72-table equation of state. Independent comparisons with numerical-electron FreeEOS at 4736 states give a maximum heat-capacity difference of 0.2952% ; 2240 states also pass thermodynamic identity checks. At the final stellar model, independent FreeEOS evaluations at all 512 actual compositions, temperatures and densities differ by at most 0.1012% in heat capacity. Interpolation uses thermodynamically consistent source states and excludes unsupported derivative stencils. These are sampled comparisons, not error bounds across the full table.

Independent matching-state comparisons between 3900 and 5500 K differ by at most 0.5986% . Lower-boundary comparisons at 4000–5600 K change the matching values by at most 0.008695% .

For further evolution, a 300-model gas-atmosphere grid extends through 6400 K. At hydrogen fractions $X = 0.15$ and 0.2 , it includes $\log_{10}[g/(\text{cm s}^{-2})] = 5.7$ and 6.0 over 5400–6400 K, for ^3He fractions of zero and 0.12 . Six independent atmospheres between these composition, temperature and gravity values differ from the tabulated matching temperatures and pressures by at most 0.4860% . Seventeen lower-boundary comparisons change matching values by at most 0.009372% . All 300 atmospheres have equation-of-state support. Equilibrium chemistry finds no condensates in the 48 structures covering the higher gravities. These calculations extend the available warm gas boundary; the plotted trajectory uses the 220-atmosphere grid described above.

The high-temperature ATOMIC opacity tables include additional mixtures with $X = 0, 0.025, 0.05$ and 0.075 at three metal fractions. Twelve independent composition comparisons on hot stellar-profile states differ by at most 0.05257% . This extends the composition coverage needed as central hydrogen is depleted; the lower-temperature tables have their own composition limits.

Dense radiative transport also depends on how the plasma changes photon propagation. An opacity extension evaluates this effect from the ATOMIC frequency spectrum and combines radiation with conduction. On the saved 3.890Tyr structure, it changes the combined opacity by at most 3.333% . Independent comparisons in the hot interior agree within 0.2871% , but comparisons in the connection to cooler tables differ by as much as 1.194% . A consistent treatment across that temperature interval is required before this extension can be used to evolve the star. Its frequency and composition sampling must cover the denser envelopes near the beginning of the track as well as the present structure. The [plasma-opacity calculations](#) give the physical assumptions and numerical comparisons.

An independent opacity comparison uses the monochromatic Opacity Project (OP) data [5]. We mix the element spectra using their tabulated ion populations and include collective electron scattering [6] and the OP screened free-free correction. At $X = 0.1$, $Z = 0.02$, $T = 9.284 \times 10^4$ K and $\rho = 0.05337 \text{ g cm}^{-3}$, the OP Rosseland mean is $1.464 \times 10^5 \text{ cm}^2 \text{ g}^{-1}$, compared with $1.497 \times 10^5 \text{ cm}^2 \text{ g}^{-1}$ from TOPS on the same frequency mesh, a difference of 2.165% . The mixture corrections lower the OP mean by 1.499% at this point.

The OP table covers 25 warm-envelope layers of the computed star, from 2.169×10^4 to 1.346×10^5 K, including all eight layers outside the MESA OPLIB mean-table range. All required corrected spectra

are positive. The family spans 53 hydrogen fractions from $X = 0$ to 0.75 at three metal fractions. Independent composition comparisons at all 18 native temperatures have a maximum difference of 0.4752%, or 0.4192% at the star’s mapped metal fraction. All 69 thermal, density and composition derivative comparisons on 23 warm-envelope layers pass. Density refinement changes their means by at most 0.001646%. These comparisons measure interpolation within the adopted atomic physics. The OP mixture omits P, Cl, K and Ti, whose combined mass fraction is 2.276×10^{-5} , and the dense core lies outside its grid. The free-free correction assumes classical electrons and ionic Debye screening. If the opacity remaining after free-electron scattering is absorption, plasma refraction raises the warm-envelope means by 0.05659–1.623%. Allowing that opacity to be either absorption or scattering gives interval widths of 0.03189–0.8259%, relative to their lower limits. These bounds retain the supplied frequency interval and do not bound missing elements or thermodynamic interpolation errors.

Eleven layers at 1.440×10^5 – 2.736×10^5 K lie between the OP table and the corrected hotter opacity. The original MESA mean tables cover all eleven. Their ordinary means differ from the TOPS low-temperature table by 1.167–3.995%, with TOPS lower. A consistent spectral plasma correction and smooth source connections remain required.

We also compare the Los Alamos OPLIB mean tables distributed with MESA [7]. At 1940 states with $X = 0$ or 0.05 and $Z = 0.01$ or 0.03, the median difference from our saved TOPS means is 2.364%, reaching 7.279%. Enlarging the original table’s density interpolation from four to six points leaves a maximum difference from TOPS of 7.197%. The plasma-cutoff setting also fails to explain the largest discrepancy. These sources use different element sets: MESA includes F, Sc, V and Co in addition to our 21 elements. On the 15 warm-envelope layers supported by both archives, OP differs from MESA by -9.730 to $+4.619\%$, after mapping the isotope number densities and interpolating in composition. These differences include atomic physics and mixture differences. A smooth connection to the molecular and high-temperature opacities, with their uncertainties assessed, is required before using this family for evolution. The [opacity comparisons](#) give the source conventions, numerical checks and physical limits.

5.1 Condensates and a hydrogen-free atmosphere

The current evolving atmosphere uses gas-only opacity. Independent chemistry checks found no condensates in the tested 3000–3200 K atmospheres, but found them in the cool upper layers of 2600–2800 K models. Effective temperature alone is therefore insufficient to decide whether grains form. Checks on the warm hydrogen-poor structures through 6400 K find no condensates; colder structures still require atmosphere calculations with grain formation and opacity included.

Grain formation can affect radiation in two ways: it removes absorbing atoms and molecules from the gas, and any suspended grains absorb and scatter light. A calculation that isolates gas depletion assumes the condensed material settles out. At solar composition and 2800 K, gas depletion changes the interior matching pressure by 0.1246% and temperature by -0.02347% . These are local gas-depletion effects, not measurements of grain opacity or of a stellar lifetime change. Dusty atmosphere models become relevant around 2800 K and below [20]. A separate grain-opacity calculation uses published optical data and explicit grain-size choices. It passes comparisons with an independent scattering program and analytic small-particle limits. Applied to a fixed gas atmosphere, it combines the optical properties with the computed alumina abundance. The maximum condensed mass fraction is 1.094×10^{-4} . For compact grains of radius $0.1 \mu\text{m}$, the integrated grain optical depths near $1 \mu\text{m}$ are 0.002610 for absorption and 0.002829 for scattering. These values describe one fixed 2800 K gas profile with suspended grains, not a recomputed atmosphere or a prediction of grain size. Absorption and scattering remain separate. Coupling these opacities to the atmospheric structure requires consistent material phases, wavelength coverage and thermal response, as described in the [grain-opacity notes](#).

Grain formation also changes thermodynamics through redistribution of atoms among molecules, ions and solids. Comparing gas-only and gas-plus-grain chemical equilibria at one test state gives a heat-capacity correction of $1.175 \times 10^5 \text{ erg g}^{-1} \text{ K}^{-1}$, compared with 2.890×10^5 from counting grains alone. A consistent thermodynamic reference, abundance normalization and treatment across phase boundaries are required before this contribution can enter the evolving model.

The atmosphere wavelength-opacity tables cover material temperatures through $1.315 \times 10^4 \text{ K}$. Nonfinite absorption near He I 4471 Å prevents use of the present source at $1.500 \times 10^4 \text{ K}$. Atmosphere calculations must remain within the valid opacity domain, with matching conditions insensitive to the chosen lower boundary.

A hydrogen-free atmosphere presents a further issue. The source programs' abundance normalization and molecular initialization currently depend on hydrogen. A direct opacity test at 4017 K shows that multiplying all number abundances by the same factor preserves composition and nearly preserves electron density, but can change individual absorption coefficients by a factor of 27.3. This one-temperature test does not measure a stellar-boundary error; it shows that a simple abundance rescaling is not yet a verified solution. The hydrogen-free limit therefore requires a source formulation that does not normalize abundances to hydrogen.

6 Energy and the cold helium remnant

The discrete energy equation used for stellar evolution is

$$L_\gamma = L_{\text{nuc,dep}} + L_{\text{grav}} - L_{\nu,\text{thermal}}, \quad L_{\text{nuc,rest}} = L_{\text{nuc,dep}} + L_{\nu,\text{nuclear}}.$$

Here L_γ is the photon luminosity, $L_{\text{nuc,dep}}$ is nuclear heat left in the star, and L_{grav} is the contribution from changing internal and gravitational energy. Nuclear and thermal neutrinos are counted separately. At 3.890Tyr in the 512-point model, in solar luminosity units,

$$L_{\text{nuc,dep}} = 0.01077, \quad L_{\text{grav}} = 3.201 \times 10^{-5}, \quad L_{\nu,\text{nuclear}} = 3.362 \times 10^{-4}, \quad L_{\nu,\text{plasma}} = 1.260 \times 10^{-7}.$$

The changing-energy term refers to the last implicit half step; the other reported diagnostics use the saved state. Before rounding, the discrete fractional luminosity residual at this saved endpoint is below 8×10^{-11} . This checks the equations implemented in the code, not an independent exact energy integral or the absence of missing physical processes.

Cold degenerate electrons have a small thermal response superimposed on a large temperature-independent contribution. Paired numerical integration around the Fermi surface retains the small response without subtracting nearly equal large terms. Pure-helium tests at 100– 10^4 K and densities 10^3 – 10^6 g cm^{-3} agree with the expected low-temperature limit to about 10^{-10} fractionally at 100 K. Electron entropy and ionic translational entropy, including mixing, are also implemented and checked against thermodynamic identities. These analytic components have not yet been combined into the production equation of state for a remnant.

Liquid and solid free energies and their derivatives are compared with Ioffe EOSFI22. Across sixteen forced-phase tests, the largest checked density-response discrepancy is 1.05×10^{-6} in ideal-ion pressure units. This tests derivative consistency within each phase; phase selection requires comparing the free energies themselves.

Helium melting requires consistent liquid and solid free energies, including quantum and mixture effects [21]. Specifying a melting value of $\Gamma = 175$ alone does not determine the latent heat or liquid heat capacity. The connection to partially ionized outer layers remains unfinished. Dense, cool atmospheres also

require nonideal chemistry and pressure-dependent absorption, including collision-induced absorption; refraction and dense-fluid effects need checking [22].

6.1 Element separation in the dense interior

Microscopic diffusion allows hydrogen, helium and metals to move relative to one another under gravity, pressure and temperature gradients, and the electric field that maintains charge balance. It can alter the hydrogen supply to burning layers after convection ceases to mix the whole star. At the center of the 3.890Tyr model, the temperature is 0.1506 times the electron Fermi temperature, so electron degeneracy must enter the transport calculation.

A separate calculation solves the coupled species equations with zero net baryonic mass flux and electric current. Numerical scattering integrals supply ion collision coefficients, while electron collisions are averaged over the finite-temperature Fermi distribution. At three layers in the saved structure, replacing the Born approximation for electron scattering with direct quantum scattering changes the total resistance to prescribed hydrogen and helium motions by at most 0.1918%. This comparison includes all 21 distinct nuclear charges of the adopted mixture, assuming fully stripped ions, and two specified screening choices. It measures the effect of that scattering approximation on the selected motions; it does not give stellar settling velocities. The forces must also agree with the equation of state. In particular, diffusion depends on how free energy changes with composition. Agreement in pressure and heat capacity alone does not test these chemical forces.

The fixed-metal comparison uses an equation-of-state representation with 152 tables at 38 hydrogen fractions and four ^3He fractions: 0, 0.03, 0.06 and 0.12. It interpolates the material free energy smoothly in both composition directions, with the ideal isotope-mixing contribution evaluated analytically. The metal inventory remains fixed. At 102 compositions spanning neutral, partly ionized and dense material, independent FreeEOS comparisons give a maximum difference of 0.1499% in the matrix of second composition derivatives. First composition derivatives agree within $2.269 \times 10^{-5} k_B/m_u$ when applied to specific free energy divided by temperature. Both source precision and composition spacing are checked independently. Across 7936 further pressure, energy and thermal comparisons, the largest heat-capacity difference is 0.1967%; at the 512 points of the saved stellar structure it is 0.1062%. All 300 warm atmosphere nodes and 48 interpolated boundary states are supported. These sampled tests establish the accuracy of the represented mixture derivatives; they do not provide derivatives for arbitrary changes in the metal abundances or a complete cold-remnant equation of state.

Composition derivatives for the species present are also evaluated when another species is exactly absent, as ^3He is in the initial model. No artificial minimum abundance is introduced. The implementation passes 128 analytic comparisons, including trace-abundance limits. At zero ^3He , five independent FreeEOS comparisons of the hydrogen curvature agree within 0.007181%. The chemical potential of an absent species has a singular ideal-mixing limit; transporting that species into an initially empty cell still requires a conservative treatment of the finite flux.

A corresponding diffusion-force calculation uses these material free-energy derivatives and imposes zero net baryonic mass flux and electric current in the species momentum equations. It passes 510 checks in 72 analytic mixtures, including trace populations and nonnegative frictional heating. These tests use synthetic collision coefficients, so they do not yet predict transport rates in the star. The force definition follows the separation of chemical and thermal driving described by Beznogov et al. [23]. When electrons are degenerate, the coefficients describing diffusion driven by temperature gradients and its associated heat transport must use the same definition. The usual classical heat-flow equations cannot simply be retained without checking this consistency [29].

The energy carried by each mass exchange is obtained from the same material free energy. At six comparison states, these enthalpy differences agree with independent FreeEOS derivatives within 0.0007647%. The coupled classical collision equations reproduce both the Burgers velocities and the full material energy flux, with nonnegative entropy production.

For electron heat transport, we compare an approximation using mean drift and one heat variable with a solution retaining the full dependence on electron energy. Both describe elastic scattering from fixed ions in a screened Coulomb potential, using finite-temperature Fermi statistics. In 422 hot layers containing 99.18% of the stellar mass, the approximation gives 27.82–81.43% of the conductivity at zero electron number current. These calculations assume fully stripped ions and compare screening by electrons alone with screening by electrons and ions. At the center, using classical statistics instead of Fermi statistics, while retaining the full energy dependence, gives only 33.27–38.07% of the corresponding conductivity. The particle density, temperature and screening length are held fixed in that comparison. Both the energy dependence and degeneracy therefore matter for this collision model.

Integrating out the electron energy dependence gives a small collision matrix for the mixture. Independent integrations recover the prescribed electron current, heat flow and dissipation, allowing these coefficients to be reused without a discretized electron-energy solve at each stellar step. Electron–electron collisions, higher-order ion recoil and correlations, and relativistic effects remain outside this comparison. These results do not measure an error in the conductivity used on the plotted stellar track. The [electron heat-transport calculation](#) gives the equations, independent checks and remaining physical requirements.

We also evaluate electron–electron scattering in the screened Born approximation, including exchange of the outgoing electrons and Pauli blocking. Its ten energy modes are combined with the electron–ion collision matrix using the actual ion abundances before reducing the mixture to its drift and heat currents. Independent solutions retaining all modes agree with this reduction; mass, current and entropy checks pass. A 54-point table of the pair collision matrix covers all 422 hot layers and both screening choices. Interpolation preserves positive dissipation and the exact momentum null mode. Eight independent stellar comparisons, withheld from the table, and three additional direct controls have a maximum relative drift/heat response difference of 0.03798%. Across the hot nonconvective region, including electron–electron collisions gives 0.5277–0.8780 times the conductivity of the same elastic-only model, evaluated at zero represented element flux. Increasing from eight to ten modes changes the interpolated stellar conductivities by at most 0.003532%. They do not measure an error in the conductivity selected for the plotted track. The [mixture collision calculation](#) gives the normalization and remaining physical approximations.

The saved structure also gives a direct measure of its temperature gradient. Writing $H_P = P/(\rho g)$ for the pressure scale height, with total pressure from the equation of state and $g = Gm/r^2$, the mass-weighted mean of $|H_P d \ln T/dr|$ is 0.1028 across the hot nonconvective interior. In its more degenerate portion, where $T/T_F < 0.3$, the mean is 0.02633. These averages cover 97.38% and 34.34% of the stellar mass, respectively. The small central gradient alone does not justify omitting thermal diffusion; its effect also depends on collision coefficients and competing forces.

Applying the coupled ion/electron comparison to the saved density, temperature and composition gradients gives provisional microscopic velocities. The calculation uses the EOS chemical derivatives and their matching material enthalpy differences, screened ion collisions and the leading ion recoil term, including the electron–electron matrix. The comparison covers 397 hot nonconvective layers containing 97.43% of the mass. The mass-weighted mean magnitude of the thermal contribution to hydrogen velocity is 23.63–24.48% of the corresponding mean without the thermal force. The range compares the two specified screening choices; it is not a physical uncertainty bound.

At enclosed mass fraction 0.6972, the provisional outward hydrogen velocity is $(1.101\text{--}1.303) \times 10^{-9} \text{ cm s}^{-1}$. The local time $H_P/|w_H|$ is 12.78–15.13 billion years; near mass fraction 0.9738 it is 3.694–4.244 billion years. These are local drift times, not remaining nuclear lifetimes. They motivate including diffusion in an evolutionary calculation. Relative to the same elastic-only model, the mass-weighted absolute change in hydrogen velocity is 2.237–2.580% of its elastic-only mean magnitude. Correlated-mixture transport, ionization, higher-order recoil and the connection between kinetic heat flow and nonideal partial enthalpies still require physical assessment. The numerical conservation and entropy checks do not establish the accuracy of these provisional velocities. The equations and retained outputs are documented in the [stellar diffusion comparison](#).

Ember’s transport equations use one shared flux at each cell face. Tests of species and energy transport conserve both quantities and increase entropy; isothermal diffusion tests decrease the Helmholtz free energy. An optional stellar interface connects species transport to hydrostatic structure, heat flow and convection, counting conductivity once. Complete radiative and partially convective numerical stellar controls close the global energy balance. Those structural controls use analytic transport coefficients, a fully ionized EOS and a grey atmosphere.

The physical ion/electron coefficients and their derivatives are evaluated at each trial state, together with the native EOS chemical potentials and exchange enthalpies. The flux agrees with an independent full species/energy matrix construction in 854 comparisons, within 6.242×10^{-14} . Its derivatives pass 576 matrix comparisons; individual responses to central trace-helium perturbations remain below numerical-difference resolution, while larger-helium controls resolve the cross terms. The finite chemical-potential terms are also checked at zero abundance.

Closed-boundary burning and diffusion controls include initially empty layers, nuclear production of an absent species and pure helium. With temperature and density fixed, independent abundance corrections are below 9.407×10^{-14} and integrated changes agree with the nuclear sources within 1.553×10^{-14} of the represented mass. No abundance floor is imposed. Heat responses and their derivatives are evaluated at fixed composition when a species is absent; eight hydrogen-free comparison states outside the collision table’s density range remain excluded. The [transport equations and comparisons](#) give the numerical tests and assumptions.

The physical transport is coupled to burning, hydrostatic structure and the energy equation in four controls on the 397-point hot interior, enclosing $0.09738 M_\odot$. Its outer pressure and temperature are held fixed, and species flux through that edge is zero. Temperature, density and composition evolve together over a million-year test interval. All four controls converge in six coupling iterations, with relative energy imbalance below 2.037×10^{-11} . These held-envelope controls exclude the cool outer layers and photosphere; partially ionized transport and consistent opacity coverage remain required for the full star.

Within a homogeneous convective region, local burning and abundance changes still require internal material redistribution. Its total outward species rate is reconstructed from the discrete continuity equation,

$$F_i = F_{\text{in}} + \sum_{j=b}^i w_j \left[S_j - \frac{X_j^{\text{new}} - X_j^{\text{old}}}{\Delta t} \right], \quad (4)$$

where w_j is the nodal mass and S_j is the nuclear abundance source. The rates at the region boundaries are retained; any regional imbalance remains an explicit residual. Subtracting the microscopic rate gives the redistribution required to maintain a homogeneous composition.

Five abundance controls use the complete 512-point structure, with its outer 115 points constrained to one mixed reservoir. Microscopic exchange is computed at the hot reservoir base. Temperature, density and the convective boundary are fixed. Across two screening choices, diffusion increases the

envelope hydrogen fraction by $0.0005974\text{--}0.0006851$ over 22 Myr. Independent corrections are below 9.128×10^{-15} in abundance, and the largest local continuity residual is 1.206×10^{-16} of stellar mass. These fixed-structure controls do not require a microscopic collision law in the cooler layers to solve the reservoir’s integrated species balance. The [convective redistribution analysis](#) gives the local rates and the limitations of this approximation.

A full-star thermal comparison carries material energy according to

$$Q_{\text{carried}} = \sum_{k=1}^2 (h_{\text{EOS},k} J_{\text{total},k} + h_{\text{kinetic},k} J_{\text{micro},k}), \quad (5)$$

where k labels the two independent species rates, $J_{\text{total},k}$ includes convective redistribution, $h_{\text{EOS},k}$ contains the material partial-enthalpy differences, and the kinetic term follows microscopic drift only. The EOS composition dependence in du/dt describes local storage, while these fluxes describe spatial transport. The rates are held fixed within each structure solve and updated in an outer iteration that also checks their heat contribution. Independent derivatives of the native hot response agree within 3.735×10^{-10} when normalized to the derivative row.

For fluid parcels in local thermal equilibrium and pressure balance, an optional radiation term adds

$$Q_{\text{rad,comp}} = \sum_k h_{\text{rad},k} (J_{\text{total},k} - J_{\text{micro},k}), \quad h_{\text{rad},k} = \frac{4P_{\text{rad}}}{\rho} \frac{(\partial P / \partial X_k)_{T,\rho}}{(\partial P / \partial \ln \rho)_{T,X}}. \quad (6)$$

The coefficient is the composition derivative of radiation enthalpy per mass at fixed temperature and total pressure. The mixing-length temperature term already uses the total EOS heat capacity. Microscopic drift remains separate from composition carried by fluid parcels. In the cool layers, omitting the microscopic term also omits its radiation subtraction. The [parcel enthalpy calculation](#) gives the derivation, independent bulk-enthalpy checks and analytic derivatives.

Direct FreeEOS queries at the exchange face, with $T = 3.641$ MK and $\rho = 13.60 \text{ g cm}^{-3}$, return ionized hydrogen to source precision and a singly ionized helium fraction of 9.334×10^{-5} . This supports the nearly fully stripped H/He approximation there within the EOS model; it does not establish the assumed metal charges or ionization of the entire envelope.

In a conditional 1 Myr calculation, all 512 points evolve in composition, temperature and structure with the nongrey atmospheric boundary. Seven coupling iterations give relative energy imbalance 5.582×10^{-10} and nuclear mass imbalance of magnitude 4.785×10^{-14} . The maximum abundance change is 3.176×10^{-5} . The test uses the retained material conductivity below 2 MK and joins to the hot kinetic calculation by 3 MK. It omits the reduced microscopic heat term below 2 MK while retaining material enthalpy transport by the total species rate. Their effects over a finite interval are tested below; these calculations are separate from the plotted track. The [whole-star heat comparison](#) records these choices and the convergence tests.

Dividing the same 1 Myr interval into two, four and eight steps gives fourteen converged substeps with the same equation and conservation tolerances. Successive endpoint abundance differences decrease from 2.359×10^{-7} to 1.206×10^{-7} and 6.138×10^{-8} , consistent with first-order time integration. The maximum relative energy imbalance is 1.473×10^{-9} . At the start, step doubling satisfies the production time-accuracy criteria with a 15.63 kyr interval: the largest structural difference is 8.227×10^{-6} in $\ln \rho$ in the deep envelope, against a target of 10^{-5} . An adaptive integration over 1 Myr uses eleven accepted intervals, increasing to as much as 0.1761 Myr. All local time-error checks pass. The accumulated hydrogen balance over its 22 implicit substeps is 1.647×10^{-17} of stellar mass. These comparisons include the initial adjustment to the chosen EOS and transport model; the physical limitations of the cooler envelope remain as stated above.

From the adaptive endpoint, a 0.1 Myr comparison with and without the parcel radiation term satisfies the time-accuracy criteria for both prescriptions, each using a full step and two half steps. The largest added radiation heat flow is 2.874×10^{-9} of surface luminosity in the convective envelope. No structural response is resolved at the numerical precision of these controls. The relative energy imbalance is below 3.664×10^{-10} . This comparison does not establish the cool microscopic heat, parcel correlations or moving-boundary treatment.

A matched 8.9 Myr comparison tests abundance errors scaled by $10^{-8} + 10^{-4}X_i$, a 10^{-4} logarithmic structure tolerance and a 10^{-3} surface-luminosity tolerance. Seven accepted intervals agree with a 43-interval control to 0.01906% in luminosity, 0.003116% in effective temperature and 0.003298% in radius. Total hydrogen differs by 1.349×10^{-8} relative, and the mixed region is identical. The accumulated local species imbalance is below 1.328×10^{-14} of stellar mass; the relative energy imbalance is below 1.828×10^{-10} . These short-interval comparisons justify the numerical settings in this regime; they do not establish a lifetime error over the complete evolution.

A continuation through 100 Myr beyond the plotted endpoint gives $T_{\text{eff}} = 5526$ K, central $X = 0.0008114$ and surface $X = 0.1762$. The convective envelope contains 2.467% of the mass. Over the last 90 Myr, 18 intervals grow to as much as 9.072 Myr; accumulated local species imbalance is 5.053×10^{-14} of stellar mass and relative energy imbalance remains below 6.220×10^{-13} .

Eight 5 Myr comparisons from this state test the envelope assumptions. Changing the intermediate opacity by $\pm 3\%$ or moving the lower temperature of the heat transition from 2 to 2.5 MK gives no resolved structural response. Removing or doubling the reduced microscopic heat term changes surface luminosity by at most 0.01320%, with the same mixed region. A prescribed internal heat flow of $\pm 0.1\%$ of surface luminosity, vanishing at the boundaries, changes surface luminosity by less than $10^{-5}\%$. This last perturbation measures sensitivity to its specified shape; it is not a bound on unknown transport. All cases pass the time-error and conservation checks. These results support the envelope approximation in the sampled regime; it must be reassessed as the convective boundary and surface composition change.

An optional closed CN cycle uses the ^{14}N proton-capture rate from Solar Fusion III, Solar Fusion II energy derivatives and the same finite-degeneracy screening as the pp network [31, 30]. Its catalyst number is the sum of the GS98 carbon and nitrogen inventories. On the 100 Myr state it adds 2.381% to the deposited pp energy; a nitrogen-only comparison gives 0.4778%. After 5 Myr of coupled evolution, the CN calculation has 0.1239% lower surface luminosity and 0.02774% lower effective temperature than the pp-only calculation. The mixed region is unchanged, and time-error, species and energy checks pass. This comparison does not restore the hydrogen consumed by carbon conversion or CN burning before its starting state; that contribution is needed for an exhaustion-age prediction. A separate 800 Myr CN sensitivity calculation from the pp comparison endpoint gives $T_{\text{eff}} = 5292$ K, central $X = 0.0005551$ and surface $X = 0.1974$. The convective envelope contains 2.371% of the mass. The last 300 Myr uses 520.7 CPU seconds, including table loading, in 32 accepted intervals. Independent time, species and energy checks pass. The core continues to warm, reaching 15.89 MK. Identifying the surface-temperature maximum requires the preceding evolution with consistent physics and fuel accounting; the decline from 5546 K can include thermal adjustment to the EOS and transport used by this continuation.

For the adopted collision model, heat carried by species motion contributes at most 0.09557% of the local luminosity on faces joining nonconvective hot cells. The conductive contribution reaches 88.27%. The pp comparison in Figure 11 includes burning, convection and slow mixing; it does not include this microscopic transport or the composition-dependent free-energy model. Further details accompany the [transport documentation](#).

A time-dependent CN network follows ^{12}C , ^{13}C and ^{14}N , including the return to carbon through proton capture by nitrogen. Coupled implicit burning, convection and thermal structure evolve these catalysts

with hydrogen and ^3He , retaining their physical mass and binding-energy changes. Fifteen coupled steps on three 512-point stellar structures take 0.2099–0.4314 s each, with relative nuclear energy imbalance at most 5.471×10^{-8} . These controls use specified initial catalyst abundances. The same network supports finite semiconvective and thermohaline mixing and microscopic transport. In the fixed-metal material comparison, carbon and nitrogen share the helium-group velocity; their mass flux enters the zero-net-mass constraint. Their total abundance varies locally, with physical helium and nuclear binding energy tracked separately from the material lookup. The metal-settling calculation instead transports them with the other heavy elements and uses the changing total metal abundance in the material properties, as described in Section 1.2. Six coupled microscopic stellar controls using the fixed-metal treatment take 1.957–2.211 s each, with maximum relative nuclear energy imbalance 1.558×10^{-7} . Over 0.2500 Myr, one full step and two half steps differ by 5.314×10^{-6} in logarithmic structure and 3.549×10^{-6} in surface luminosity.

Evolving the 10 Gyr initial structure with the time-dependent CN network, Solar Fusion III rates and the refined equation of state gives $T_{\text{eff}} = 3062$ K, $T_c = 6.377$ MK and $X = 0.3598$ at 2.500Tyr. The star remains fully convective. The initial catalyst composition takes carbon as ^{12}C and nitrogen as ^{14}N ; ^{13}C is produced by burning. By 2.500Tyr, 1.458×10^{-4} of the initial carbon-12 has been destroyed, while only 2.976×10^{-8} of the carbon has become nitrogen.

In the fixed-metal comparison, full convection is lost near 3.549Tyr, at $T_{\text{eff}} = 3230$ K and central $X = 0.1744$. An interior radiative layer separates the convective core from the envelope. Carbon-12 depletion is 0.02652, with 0.02524 accumulated as carbon-13 and net elemental conversion to nitrogen of 0.001275. The two proton captures therefore cannot be represented as instantaneous conversion of carbon to nitrogen during this phase. The central convective region subsequently disappears near 3.551Tyr; the envelope remains convective. Independent time, species and energy checks pass through both transitions. The [loss of full convection](#) and [radiative-center transition](#) are recorded with their saved age brackets.

The fully convective calculation includes material conduction and heat carried by convective composition redistribution. Comparisons with the full adopted microscopic heat model at 2.500Tyr and 3.300Tyr give no resolved structural or luminosity change over 10 Myr, with the same convective regions and passing fuel and energy checks. This finite-interval comparison does not bound the heat contribution after a radiative region develops. For the denser cool envelopes we use TOPS opacity; the OP table used for the hotter stellar model does not cover those densities. The material and screening calculations retain the fixed-GS98 mixture approximation. The [CN calculation](#) records the numerical and physical scope.

A comparison initialized with unprocessed catalysts at 2.500Tyr loses full convection about 1.758 Gyr later, a 0.04952% age difference. The comparison also includes different timestep histories. A structure during this transition has a mainly radiative layer at enclosed mass fractions 0.1770–0.2391, separating a convective core from the outer convective envelope. Conduction supplies 4.4–4.8% of its diffusive heat flow, and species motion carries about 0.1% of the local luminosity. Conduction thus matters for the convection threshold even though radiation carries most of the heat. Removing conductivity on this fixed structure would make the layer unstable under the Ledoux criterion; this diagnostic does not specify the evolution of a star without conduction. The [transport profile](#) retains the composition and heat fluxes of the converged stellar step.

Independent source regions and stellar layers are evaluated by reusable threads sharing one set of material tables. Physical sums retain their specified order, and exact collision responses are reused between evaluations. A matched full-step/two-half-step comparison gives identical physical outputs and uses 16.07% less CPU than creating new evaluation threads each time.

The collision integrals use 96 Jacobi and 128 Legendre points. Comparing against 384/512 points over 1076 valid states gives maximum scaled coefficient and derivative differences of 4.851×10^{-9} and 3.645×10^{-7} . Two consecutive 25 Myr full-step/two-half-step stellar comparisons preserve the convective regions and use 38.20% less solver CPU. Electron matrices depend on two local variables; applying the chain rule to these two partials supplies all six physical derivatives. Reusing this calculation and omitting only exactly zero spline terms saves another 7.829% of CPU in a two-thread, 50 Myr comparison, with unchanged physical prescriptions and passing conservation checks.

The fixed-metal comparison approximates its metal contribution to collisions with two representative carriers in place of the 19 GS98 metal species. Their number-weighted charge moments through cubic order, total mass and mass-weighted charge match the mixture. The same approximation enters static ion screening; the EOS and nuclear abundances retain the physical metal inventory. A 500 Myr comparison with all metal collision carriers gives a luminosity change of 0.00006121% and a maximum absolute species difference of 8.868×10^{-8} , with the same convective regions and passing time and conservation checks. Solver CPU decreases by 36.93% in that comparison.

The relative composition tolerance of 0.001 is supported by a separate 700.4 Myr comparison: luminosity differs by 0.001256%, with unchanged convective boundaries. The full-step/two-half-step, fuel and energy checks remain active. These finite-interval controls measure numerical accuracy and computing cost; they do not establish a full-lifetime error bound.

The gas atmosphere table contains 354 usable source states, including $X = 0.70$ through 4000 K. Direct source calculations at $X = 0.575$ and 3500/3700 K differ from interpolation by 1.113%/1.837% in matching temperature. Applying their paired temperature and pressure differences in 100 Myr stellar comparisons changes luminosity by 0.5031%/0.7126%. The second comparison shifts the convective boundary by one mass cell, 0.3116% of the stellar mass. We retain this measured local uncertainty; these comparisons do not bound the error over the full trajectory.

6.2 Carbon conversion and molecular opacity

The small metal mass fraction does not ensure that carbon conversion has only a small opacity effect. CO formation couples the available carbon and oxygen; changing their relative abundances can change the molecules that absorb radiation [39]. Complete conversion of the initial GS98 carbon to nitrogen captures a hydrogen mass fraction of 5.728×10^{-4} and raises the metal mass fraction by the same amount. An EOS approximation must account for that fuel and energy without misidentifying the physical helium abundance. A calculation of core exhaustion must also follow catalyst mixing during and after full convection.

We compare the GS98 material opacity with compositions in which half or all of the initial carbon has become nitrogen. The final hydrogen abundance is $X = 0.20$, with $X_3 = 0$, in every case. The converted mixtures include the captured proton mass in nitrogen; helium is adjusted to preserve the total baryonic mass, and the other metal nuclei are unchanged. The source solves chemical equilibrium and wavelength-dependent absorption at the same temperature and density. At $\rho = 5.995 \times 10^{-6} \text{ g cm}^{-3}$, the ratios of Rosseland absorption means to the fixed-GS98 reference are:

T (K)	Half the carbon converted	All the carbon converted
2705	1.225	1.407
3521	1.151	1.263
4583	0.9971	0.9912
5964	0.9901	0.9744

These means cover wavelengths from 0.09 to $30\mu\text{m}$ and exclude scattering and grains. Across the sampled densities between 10^{-8} and 10^{-4}g cm^{-3} , half conversion raises the 2705 K mean by 18.67–30.22%. Thus a fixed metal mixture requires a measured atmosphere comparison before it can support evolution with changing carbon and nitrogen abundances. These are material-state sensitivities; they do not directly predict a change in stellar effective temperature.

A coupled gas-atmosphere comparison uses $X = 0.20$, $X_3 = 0$, $T_{\text{eff}} = 3200\text{ K}$ and $\log g = 5.15$. With half the carbon converted, the temperature at optical depth 100 increases by 1.601%, gas pressure falls by 5.681% and density falls by 7.877%. The atmosphere includes Rayleigh scattering and omits condensates. Quarter conversion gives temperature and pressure changes of +0.8586% and -3.147% at the same surface state. At 5400 K, half conversion instead changes them by -0.005640% and $+0.7818\%$. Thus the response depends on temperature as well as the amount of carbon converted.

Applying the measured cool-atmosphere temperature and pressure factors together to the 3.400Tyr structure changes its surface luminosity by -5.162% and effective temperature by -1.361% after 10 Myr. Applying the warm-atmosphere factors to the late structure changes these quantities by $+0.4487\%$ and $+0.1209\%$ after 0.2500 Myr. These constant boundary perturbations establish a material stellar response, particularly in the cool model; they do not provide a lifetime correction.

The warm material-opacity tables also extend to $X = 0.30$, at $X_3 = 0$ and 0.12, with 16 temperatures from 1822 to $1.315 \times 10^4\text{ K}$ and 19 densities from 10^{-13} to 10^{-3}g cm^{-3} . Both compositions pass the source-table checks. Twelve coupled atmosphere solutions at 5200, 5400 and 5600 K cover $\log g = 5.15$ and 5.4 at both helium-3 fractions. Direct solutions at $X = 0.25$ differ from interpolation between $X = 0.20$ and 0.30 by about 0.97% in matching-layer pressure and 0.08% in temperature. Applying the paired temperature and pressure correction to the late stellar structure changes luminosity by -0.3997% and effective temperature by -0.1082% after 0.2500 Myr. These two composition comparisons and the finite stellar interval characterize the tested part of the atmosphere grid. The lowest-hydrogen tested envelope has surface $X = 0.1974$, near the composition boundary of this atmosphere family.

7 Hydrogen exhaustion and remnant cooling

Extending the track through hydrogen exhaustion requires atmosphere and interior inputs covering the changing composition, temperature and density, including hydrogen-free mixtures and grain formation where relevant.

As composition becomes stratified, conservative microscopic diffusion and settling must include the electric field that maintains charge balance and the associated energy exchange. The importance of omitted nuclear reactions, thermal-neutrino processes and semiconvective heat transport must be assessed in the conditions actually reached.

Hydrogen exhaustion and cooling require distinct measurements. Central hydrogen crosses $X = 10^{-3}$, 10^{-4} and 10^{-5} near 3.650Tyr, 3.660Tyr and 3.767Tyr, respectively. At the first crossing, the star retains about $0.0147M_{\odot}$ of hydrogen. We will also record when nuclear heat falls below 10%, 1% and 0.1% of photon luminosity during cooling, including any later recovery. Remaining hydrogen mass and shell burning must be followed; a depleted center alone does not establish that burning has ended everywhere.

A consistent liquid/solid helium equation of state, conductive transport and a dense atmosphere must allow the star to evolve through 1000, 500 and 100 K on the descending-temperature branch. The resulting cooling ages depend on the thermal reservoir, phase changes and atmospheric boundary.

The present calculation assumes fixed mass and no external heating. External heating, stellar encounters and particle decay are treated as conditional extensions of the isolated track [16]. The timeline in

Section 9 includes these alternatives, cooling past 100 K, loss of degeneracy during mass depletion, and the end of the bound remnant. For close encounters, energy transferred into stellar oscillations is distinct from retained heat. Gravitational radiation can remove mode energy [17]; competing physical damping mechanisms must determine the thermalized fraction before it enters a remnant-heating history.

8 Computing time and reproducibility

Reading the 152 equation-of-state tables is a separate startup cost. Buffered numeric parsing reduces a matched load from 233.5 to 64.75 s, a factor of 3.606, preserving the tested coefficients and stellar solutions exactly. All subsequent stellar steps share those loaded tables.

The matched full-star diffusion comparison described above takes 82.83 s of solver time, compared with 667.4 s for the stricter control: a factor of 8.058. There are 21 and 129 implicit solves, respectively, including the full-step and two-half-step comparisons, and no rejected intervals. Elapsed times including one EOS table load are 127.1 and 718.2 s. The largest accepted interval increases from 0.2460 to 2.782 Myr. Both calculations have 512 mass points and the same physical inputs.

The abundance tolerance for the outer composition/structure iteration is 10^{-10} , with 10^{-11} inside the species solve; independent energy and species-conservation tests remain satisfied. The local time-error criteria are distinct from these nonlinear-solver tolerances. A safeguarded Newton inversion of electron density replaces repeated bisection in the collision calculation, using the same quadrature and table domain. Among 903 scalar and 903 derivative queries, scalar responses agree within 1.761×10^{-14} and physically scaled derivatives within 1.552×10^{-14} . The scalar calculation is 2.064 times faster and the derivative calculation 1.356 times faster in alternating measurements.

A separate late 2 Gyr comparison increases the structure time-error allowance from 0.0003 to 0.001 with the physical tables and conservation checks unchanged. Solver CPU falls by 65.47%, while endpoint luminosity changes by 1.078%, radius by 0.4062% and temperature by 3.114 K. These changes exceed the stated comparison limits, so the plotted track retains the original setting.

The short comparison does not determine the runtime of the remaining hydrogen burning or remnant cooling. The [accuracy and timing report](#) supplies the full comparison and physical scope.

The initial diffusion coefficients give an explicit stability limit of 672.7–936.1 years. Among the last 500 saved structures, steps lasting at least one million years have a median duration of 59.44 million years. This comparison motivates implicit transport; convergence does not determine the time accuracy of a stellar step.

Computing the model sequences shown in Figure 1 used 47.49 CPU-hours: 36.31 hours for the metal-settling trajectory, 10.98 hours for the fixed-metal comparison, and 2.259 hours for the pp comparison. The first two share 861 states and 2.052 CPU-hours, counted once in the combined total. The displayed table-building work is about 2.6 times larger in CPU-hours. These sequence totals include native table loading, time-step comparisons and adaptive retries within the runs supplying the plotted curves; independent controls and discarded tracks are excluded. Their rough 3×10^{14} FLOP estimate assumes 2×10^9 operations per CPU-second, with an estimated range $0.6\text{--}6 \times 10^9$. This is a broad conversion informed by CPU arithmetic benchmarks, rather than a measured whole-program operation rate. Both costs use the same unit: one CPU-hour is one processor core working for one hour. Sequence costs are measured process CPU time. Table costs are estimated from serial job durations and, for some EOS batches, elapsed time multiplied by the number of workers. Documented sleep is removed; remaining waiting and unused batch capacity may overestimate CPU use.

Preparing physical tables is separate from integrating a stellar track. For the 152-table equation of

state described above, an alternating timing comparison over the saved 512-point profile gives 0.3561 microseconds per evaluation with smooth composition interpolation, compared with 0.1469 microseconds using linear interpolation of the same tables. The test includes temperature and density responses, with repeated access to cached data; it does not measure a stellar time step or the full diffusion calculation. A standalone probe uses at most 4.149 GB of memory. Table preparation and loading are reusable costs, distinct from evaluation at each stellar mass point.

Repeated nuclear calculations are an identified production cost. Enlarging the cache of exactly matching stellar states reduced CPU time by 42.21% in a short alternating comparison. Sharing the electron calculation within each nuclear response saved a further 14.97% in a separate short comparison with the enlarged cache. All full output files in each comparison agree exactly; 2048 saved late stellar states also give identical nuclear responses and derivatives with the additional electron reuse. Both forms of reuse are included in Ember. A fixed-input timing comparison through 3.757Tyr isolates the larger cache with identical physical inputs and entire output files: CPU time falls from 162.4 to 101.8 minutes, a reduction of 37.31%. The additional electron reuse was not part of that pair. Concurrent background load varied, so this single long pair is not an isolated-machine timing guarantee. The savings during shell burning and remnant cooling remain unmeasured.

The unmodified F77 AJR run used 27.37 CPU seconds through its programmed cooling stop. Ember’s finer mass mesh, coupled implicit solves, time-step checks and different physics perform additional work. Their separate contributions to the full speed difference have not been measured. An atmosphere replay spent about 54% of its measured time in chemistry and only 0.0033 seconds in the dense matrix solve; that describes one atmosphere calculation, not the stellar integrator.

The [GitHub repository](#) contains code, table-generation instructions, small records of input choices and checks, this paper and the numerical data needed for its figures. The LBA97 comparison records the selected image coordinates, axis calibration and original PDF checksum; its curves can be rebuilt without the original image. The F77 figure data are extracted from accepted models and checked against the recorded output hashes. Full raw outputs, generated physical tables, saved stellar states and executable archives remain local. Instructions for recreating the figures and recovering the recorded stellar calculation accompany the [paper files](#).

The repository supplies figure data, source specifications and validation records; regenerating the bulk physical tables is required for stellar calculations from a fresh checkout.

References

- [1] Jermyn, A. S., et al. (2023). Modules for Experiments in Stellar Astrophysics (MESA): Time-Dependent Convection, Energy Conservation, Automatic Differentiation, and Infrastructure. *ApJS*, 265, 15. <https://arxiv.org/abs/2208.03651>.
- [2] MESA documentation. Table atmospheres. <https://docs.mesastar.org/en/latest/atm/table.html>.
- [3] Rohrmann, R. D., Althaus, L. G., & Kepler, S. O. (2011). Lyman α wing absorption in cool white dwarf stars. *MNRAS*, 411, 781–791. doi:10.1111/j.1365-2966.2010.17716.x.
- [4] Rohrmann, R. D., Althaus, L. G., García-Berro, E., Córscico, A. H., & Miller Bertolami, M. M. (2012). Outer boundary conditions for evolving cool white dwarfs. [arXiv:1209.2452](https://arxiv.org/abs/1209.2452).
- [5] Seaton, M. J. (2005). OP data on CD for mean opacities and radiative accelerations. doi:10.1111/j.1365-2966.2005.00019.x.
- [6] Boercker, D. B. (1987). Collective effects on Thomson scattering in the solar interior. *ApJ*, 316, L95–L98. [Published article](#).

- [7] Farag, E., et al. (2024). An Expanded Set of Los Alamos OPLIB Tables in MESA: Type-1 Rosseland-mean Opacities and Solar Models. *The Astrophysical Journal*. doi:10.3847/1538-4357/ad4355.
- [8] Colgan, J., et al. (2016). A New Generation of Los Alamos Opacity Tables. *ApJ*, 817, 116. doi:10.3847/0004-637X/817/2/116.
- [9] Los Alamos National Laboratory. TOPS opacity service, ATOMIC/New elemental library and mixture queries. aphysics2.lanl.gov/opacity/lanl/.
- [10] Davis, Y. T., et al. (2024). The EBLM Project XII. An eccentric, long-period eclipsing binary with a companion near the hydrogen-burning limit. [arXiv:2312.09156v2](https://arxiv.org/abs/2312.09156v2), Table 2 and abstract.
- [11] Agol, E., et al. (2021). Refining the Transit-timing and Photometric Analysis of TRAPPIST-1: Masses, Radii, Densities, Dynamics, and Ephemerides. [arXiv:2010.01074](https://arxiv.org/abs/2010.01074), Table 7.
- [12] Ribas, I., et al. (2017). The full spectral radiative properties of Proxima Centauri. [arXiv:1704.08449](https://arxiv.org/abs/1704.08449), Table 11.
- [13] Laughlin, G., Bodenheimer, P., & Adams, F. C. (1997). The End of the Main Sequence. *ApJ*, 482, 420. doi:10.1086/304125.
- [14] Chabrier, G., & Baraffe, I. (1997). Structure and evolution of low-mass stars. *A&A*, 327, 1039. [arXiv:astro-ph/9704118](https://arxiv.org/abs/astro-ph/9704118), Table 2.
- [15] Alexander, D. R., Johnson, H. R., & Rypma, R. L. (1983). *ApJ*, 272, 773, Table 2. [Published opacity table](#).
- [16] Adams, F. C., & Laughlin, G. (1997). A dying universe: the long-term fate and evolution of astrophysical objects. *Rev. Mod. Phys.*, 69, 337. doi:10.1103/RevModPhys.69.337.
- [17] Rathore, Y., Blandford, R. D., & Broderick, A. E. (2005). Resonant excitation of white dwarf oscillations in compact object binaries—I. The no back reaction approximation. *MNRAS*, 357, 834. [Published calculation](#).
- [18] Baraffe, I., Chabrier, G., Allard, F., & Hauschildt, P. H. (1998). Evolutionary models for solar metallicity low-mass stars: mass–magnitude relationships and color–magnitude diagrams. *A&A*, 337, 403. [arXiv:astro-ph/9805009](https://arxiv.org/abs/astro-ph/9805009).
- [19] Haft, M., Raffelt, G., & Weiss, A. (1994). Standard and nonstandard plasma neutrino emission revisited. [arXiv:astro-ph/9309014](https://arxiv.org/abs/astro-ph/9309014).
- [20] Chabrier, G., Baraffe, I., Allard, F., & Hauschildt, P. (2000). Evolutionary models for very low-mass stars and brown dwarfs with dusty atmospheres. *ApJ*, 542, 464. [arXiv:astro-ph/0005557](https://arxiv.org/abs/astro-ph/0005557).
- [21] Baiko, D. A., & Chugunov, A. I. (2022). *Ab initio* thermodynamics of one-component plasma for astrophysics of white dwarfs and neutron stars. [arXiv:2112.04822](https://arxiv.org/abs/2112.04822).
- [22] Blouin, S., Dufour, P., & Allard, N. F. (2018). A new generation of cool white dwarf atmosphere models. I. Theoretical framework and applications to DZ stars. [arXiv:1807.06616](https://arxiv.org/abs/1807.06616).
- [23] Beznogov, M. V., Potekhin, A. Y., & Yakovlev, D. G. (2016). Diffusive heat blanketing envelopes of neutron stars. *MNRAS*, 459, 1569. doi:10.1093/mnras/stw751.
- [24] Thoul, A. A., Bahcall, J. N., & Loeb, A. (1994). Element diffusion in the solar interior. *ApJ*, 421, 828. [arXiv:astro-ph/9304005](https://arxiv.org/abs/astro-ph/9304005).
- [25] Denissenkov, P. A., Herwig, F., Bildsten, L., & Paxton, B. (2013). MESA Models of Classical Nova Outbursts: The Multicycle Evolution and Effects of Convective Boundary Mixing. *ApJ*, 762, 8. [Published article](#).
- [26] Bauer, E. B., & Bildsten, L. (2019). Polluted White Dwarfs: Mixing Regions and Diffusion Timescales. *ApJ*, 872, 96. [arXiv:1812.09602](https://arxiv.org/abs/1812.09602).
- [27] Charbonnel, C., & Zahn, J.-P. (2007). Thermohaline mixing: A physical mechanism governing the photospheric composition of low-mass giants. *A&A*, 467, L15. [arXiv:astro-ph/0703302](https://arxiv.org/abs/astro-ph/0703302).

- [28] Paxton, B., et al. (2011). Modules for Experiments in Stellar Astrophysics (MESA). *ApJS*, 192, 3. [Published methods](#).
- [29] Paxton, B., et al. (2018). Modules for Experiments in Stellar Astrophysics (MESA): Convective Boundaries, Element Diffusion, and Massive Star Explosions. [arXiv:1710.08424](#), section 3.2.2.
- [30] Adelberger, E. G., et al. (2011). Solar fusion cross sections. II. The pp chain and CNO cycles. *Reviews of Modern Physics*, 83, 195.
- [31] Acharya, B., et al. (2025). Solar fusion III: New data and theory for hydrogen-burning stars. *Rev. Mod. Phys.*, 97, 035002. [doi:10.1103/8lm7-gs18](#).
- [32] Baraffe, I., Homeier, D., Allard, F., & Chabrier, G. (2015). New evolutionary models for pre-main sequence and main sequence low-mass stars down to the hydrogen-burning limit. [arXiv:1503.04107](#).
- [33] Takenaka, A., et al. (Super-Kamiokande Collaboration, 2020). Search for proton decay via $p \rightarrow e^+ \pi^0$ and $p \rightarrow \mu^+ \pi^0$ with an enlarged fiducial volume in Super-Kamiokande I–IV. [arXiv:2010.16098](#).
- [34] Fukugita, M., & Yanagida, T. (1986). Baryogenesis without grand unification. *Phys. Lett. B*, 174, 45. [doi:10.1016/0370-2693\(86\)91126-3](#).
- [35] Adams, F. C., Laughlin, G., Mbonye, M., & Perry, M. J. (1998). The gravitational demise of cold degenerate stars. *Phys. Rev. D*, 58, 083003. [arXiv:astro-ph/9808250](#).
- [36] Adams, F. C., Kane, G. L., Mbonye, M., & Perry, M. J. (2001). Proton decay, black holes, and large extra dimensions. [arXiv:hep-ph/0009154](#).
- [37] Bell, N. F., et al. (2024). Heavy Dark Matter in White Dwarfs: Multiple-Scattering Capture and Thermalization. [arXiv:2404.16272](#).
- [38] Gasques, L. R., et al. (2005). Nuclear fusion in dense matter: Reaction rate and carbon burning. *Phys. Rev. C*, 72, 025806. [arXiv:astro-ph/0506386](#).
- [39] Lederer, M. T., & Aringer, B. (2009). Low temperature Rosseland opacities with varied abundances of carbon and nitrogen. [Published article](#); [arXiv:0810.5672](#).
- [40] Potekhin, A. Y., & Chabrier, G. (2012). Thermonuclear fusion in dense stars: Electron screening, conductive cooling, and magnetic field effects. *A&A*, 538, A115. [arXiv:1201.2133](#).
- [41] Chugunov, A. I., & DeWitt, H. E. (2009). Nuclear fusion reaction rates for strongly coupled ionic mixtures. *Physical Review C*, 80, 014611. [arXiv:0905.3844](#).

9 Lifetime timeline

The timeline follows formation, hydrogen burning, remnant cooling and the conditional disappearance of the remnant. The metal-settling calculation reaches 4.002Tyr; its numerical milestones below come from saved structures. Initial-model and helium-3 comparison values are identified separately. Formation precedes age zero. Cooling and environmental ages remain to be calculated, and several processes can overlap. Disappearance means loss of the baryonic remnant through decay products.

9.1 Formation and hydrogen burning

Age or condition	Event	Physical description and basis
Before $t = 0$	Formation and accretion	Cloud collapse assembles the protostar. Accretion, a disk and any outflows set its initial mass, entropy and angular momentum. These stages are outside the present fixed-mass calculation.
Before $t = 0$	Deuterium burning and contraction	The young object contracts toward sustained hydrogen burning; deuterium is consumed, and lithium is depleted as the interior heats. Low-mass pre-main-sequence models supply the comparison, rather than an Ember formation age [32].
$t = 0$	Fully convective hydrogen-burning model	The pp comparison begins with $X = 0.7$, $Z = 0.02$, $R = 0.1259R_{\odot}$ and $T_{\text{eff}} = 2766$ K. The pp chain converts hydrogen to helium while convection mixes the star.
0.7103Tyr	Maximum ${}^3\text{He}$ abundance	In the pp comparison, the central ${}^3\text{He}$ mass fraction reaches 0.1003 and then decreases as its destruction catches up with its production. LBA97 gives 0.0995 at 1.38Tyr.
3.549Tyr	Loss of full convection	A radiative layer separates the convective center from the envelope; central $X = 0.1744$. The center becomes radiative near 3.551Tyr. LBA97 reports a central radiative core at 5.742Tyr and $X = 0.16$.
3.560Tyr	Separate core and envelope composition	The convective mass fraction is 0.5736. Central $X = 0.1265$ differs from surface $X = 0.1853$; whole-star mixing has ended.
3.650–3.767Tyr	Central hydrogen depletion	Central X crosses 10^{-3} , 10^{-4} and 10^{-5} near 3.650, 3.660 and 3.767Tyr, respectively. Hydrogen remains in the envelope and burns outside the depleted center.
3.838Tyr	Local surface-temperature maximum	Surface temperature reaches 4754 K, then declines by 141.5 K over 164.3 Gyr. Residual hydrogen burning still dominates luminosity; this maximum does not by itself establish a passive cooling sequence.
4.002Tyr	Computed metal-settling endpoint	$T_{\text{eff}} = 4612$ K and central $X = 3.528 \times 10^{-8}$. Conduction and radiation carry heat through the core; the convective envelope contains 1.553% of the mass.
Age to be calculated	Sustained white-dwarf cooling	Follow the decline of nuclear power and thermal energy as the helium remnant cools. LBA97 reaches 5807 K; this is a comparison value, not an imposed Ember temperature.

9.2 Cooling and environmental exposure

These stages are identified by local conditions or their effect on the energy balance. A surface temperature of 100 K is an intermediate milestone; it does not imply a 100 K core. The calculation will compare the following processes before assigning their event ages.

Condition	Event	Physical description and calculation
Declining nuclear power	Residual hydrogen burning fades	Record nuclear heating below 10%, 1% and 0.1% of photon luminosity, including any later recovery. Retain the envelope fuel supply and thermal energy rather than assuming an immediately passive remnant.
Transport competes with evolution	Diffusion and separation	Hydrogen can collect at the surface while heavier species settle. The atmosphere's composition follows transport and mixing; a helium core does not imply a helium atmosphere.
Envelope cools	Molecules, grains and dense atmospheres	Follow molecular absorption, collision-induced absorption, scattering, condensation and settling where the local conditions permit them. Couple opacity to composition and energy transport [20, 22].
Interior reaches a phase boundary	Crystallization and chemical separation	Evaluate helium and mixture phase equilibria, latent heat, separation energy and solid heat capacity. The front's progress is determined by density and core temperature, not one assigned surface temperature [21].
Convection reaches deeper layers	Envelope and core become thermally coupled	Determine whether and when the convective envelope reaches the nearly isothermal interior, changing the relation between stored heat and luminosity. Its order relative to crystallization is to be calculated.
$T_{\text{eff}} = 1000, 500, 100 \text{ K}$	Cold-remnant milestones	Record ages, core temperature, radius, composition, solid fraction and power sources on the cooling branch. Below 100 K, continue with the appropriate solid and partly neutral matter description.
Including ages $\gtrsim 10^{14} \text{ yr}$	Changing environment and accretion	Compare retained and ejected objects in evolving stellar and dark-matter environments. Gas, debris and interstellar solid objects provide conditional accretion and heating; no unique late encounter rate is assumed.
Close passages	Tidal heating, capture or collision	For surviving unbound WD pairs, distinguish heat from spin and persistent oscillations. Couple an encounter survey to waiting-time distributions and cooling after each pulse; captures and collisions form separate outcomes.
Allowed particle interactions	Dark-matter heating	Compare capture, thermalization, evaporation and possible annihilation with the photon luminosity. Include no annihilation heating as an alternative; crystallization can change thermalization [37].
Nuclear times become competitive	Density-driven fusion	Estimate pycnuclear and impurity reactions against cooling and decay times, using the actual helium-rich composition. Carbon-burning results do not directly determine helium-remnant rates [38].

9.3 Decay and disappearance: conditional branches

Proton decay has not been detected. The Super-Kamiokande limit for $p \rightarrow e^+ \pi^0$ is a partial lifetime $\tau_p/B > 2.4 \times 10^{34}$ yr at 90% confidence [33]. Baryogenesis does not itself require ordinary proton decay; leptogenesis provides a counterexample [34].

Virtual-black-hole interactions supply another possible decay channel. A Planck-scale estimate is $\tau_p \sim 10^{45}$ yr, with rate set by the assumed quantum-gravity interaction [36]. This differs from collective tunneling of stellar matter through its degeneracy barrier, which can be far more suppressed [35]. We therefore consider finite-lifetime scenarios and a stable-baryon branch rather than assigning one disappearance date. The assumed channel must specify which bound and free nucleons decay.

Condition	Event	Physical description and calculation
Decay heating exceeds other sources	Decay-supported luminosity	Compare deposited decay energy with cooling and environmental power. This can occur long before a substantial fraction of the mass is lost. Escape fractions depend on decay products and the evolving column density.
Accumulated decays	Composition changes and mass loss	Track baryonic mass, nuclear daughter products, subsequent decays and reaction energy. Begin with coarse composition groups and refine only where they change pressure, retained heat or phase order.
Pressure support changes	Loss of degeneracy	Follow the transition from a degenerate remnant to matter whose structure depends on atomic and molecular forces. Recalculate the mass-radius relation; do not impose a transition mass from a different composition.
Columns become small	Decay products and thermal radiation escape	Separate transparency to energetic decay products from transparency at thermal wavelengths. Decreasing energy retention changes luminosity and temperature; an optically thin object need not radiate as a blackbody.
Binding becomes insufficient	End of the bound remnant	Allow fragmentation or dispersal if the material and energy balance require it. This is distinct from decay of every formerly associated baryon. Accretion, capture or collision can interrupt the isolated path at other stages.
Few baryons remain	Discrete survival and final decay	Replace a continuous mass variable with particle survival probabilities. Specify any stable products. If all original baryons have finite lifetimes, their last decay is a random event; an exponential mean never reaches zero at a finite time.
Stable-baryon alternative	No decay-imposed endpoint	Continue cooling, nuclear and environmental alternatives without assigning a proton-decay date. Any gravitational destruction channel requires its own assumptions and rate.

The degree of physical detail follows the quantity being predicted. Resolved atmospheres and interior transport determine the stellar track and cooling ages. At more uncertain stages, thermal balances, encounter distributions and conditional decay models are sufficient unless additional structure changes the result. Numerical errors must still be smaller than the signal of interest, especially for weak tidal heating. Successive descriptions will be compared where both apply, preserving mass, composition and stored energy.

9.4 A reference clock for the decay era

A useful analytic comparison assumes that every baryon has the same constant, independent mean lifetime τ_B , with no accretion or stable daughter baryons. Let t_d measure elapsed time from the start of this comparison. Then

$$\langle M_B(t_d) \rangle = M_{B,0} e^{-t_d/\tau_B}, \quad L_{\text{heat}} = f_{\text{dep}}(t_d) \frac{\langle M_B \rangle c^2}{\tau_B}. \quad (7)$$

The second expression treats baryon mass as the available rest energy; a physical channel also accounts for daughter masses and nuclear binding. The deposited fraction f_{dep} is allowed to evolve. This is a reference mass and energy model, not a resolved calculation of the late atmosphere.

Expected initial baryonic mass lost	t_d/τ_B
50%	0.6931
90%	2.303
99%	4.605
99.99%	9.210

For N_0 independently decaying baryons the last-event distribution is

$$\Pr(t_{\text{last}} \leq t_d) = \left(1 - e^{-t_d/\tau_B}\right)^{N_0}, \quad \langle t_{\text{last}} \rangle = \tau_B H_{N_0}, \quad (8)$$

where $H_N = \sum_{k=1}^N 1/k$. For an initial baryonic mass $0.1M_\odot$, $N_0 \simeq 1.2 \times 10^{56}$. The mean last decay occurs near $129.7\tau_B$, with a central 90% interval from 128 to $132.1\tau_B$. This follows all tagged original baryons, including any dispersed material; it does not date the survival of a bound star. Correlated processes, composition-dependent lifetimes and stable products change this reference.

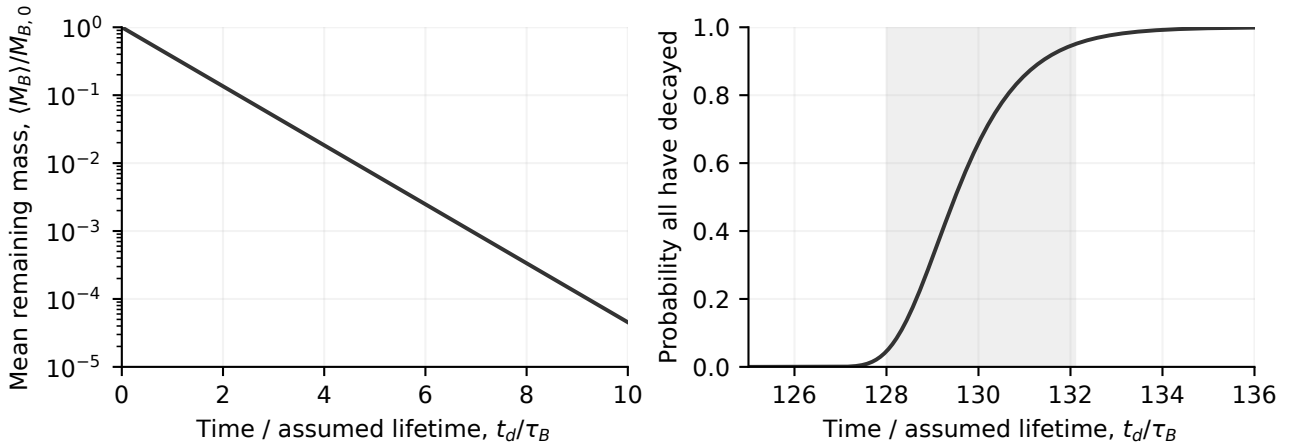


Figure 15: Conditional decay reference for an initially $0.1M_\odot$ baryon population with one independent lifetime. Left: expected remaining mass. Right: probability that all original baryons have decayed; shading marks the central 90% last-event interval. The two time ranges differ. These curves are analytic consequences of the stated assumptions, not an extrapolation of the computed Ember track.